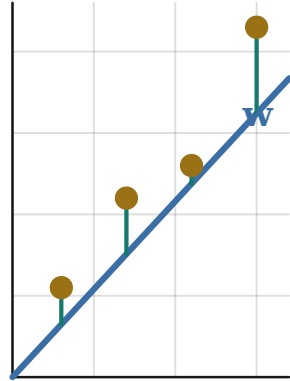


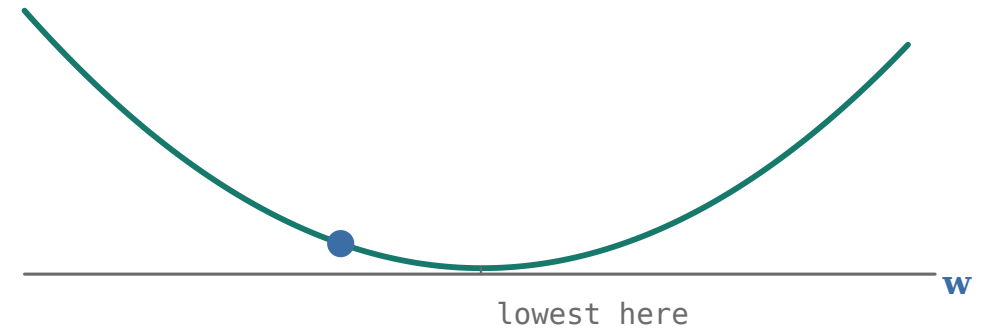
From Derivatives to Learning

A model that improves from data

data, fixed



$L(w)$



$$\hat{y} = f_{\theta}(x), \quad L_1(\theta) = \frac{1}{2} (f_{\theta}(x) - y^*)^2$$

x, y^* given; the update never moves them

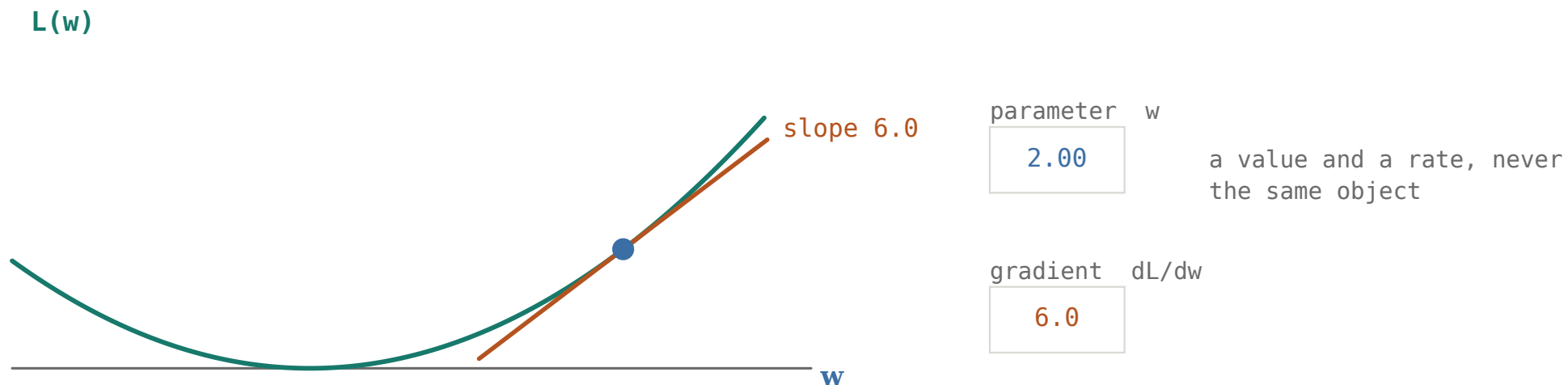
θ the only thing we may change

L_1 one example; the curve sums all four

Which way should θ move, and how far?

From Derivatives to Learning

What the gradient gives us



$$g_j = \frac{\partial L}{\partial \theta_j}, \quad g = \nabla_{\theta} L \in \mathbb{R}^p$$

g_j one number per parameter
 g a column, shaped like θ

θ_j the parameter it belongs to
 p how many parameters there are

From Derivatives to Learning

Which quantity changes?

x

changed?

y^*

changed?

θ

changed?

η

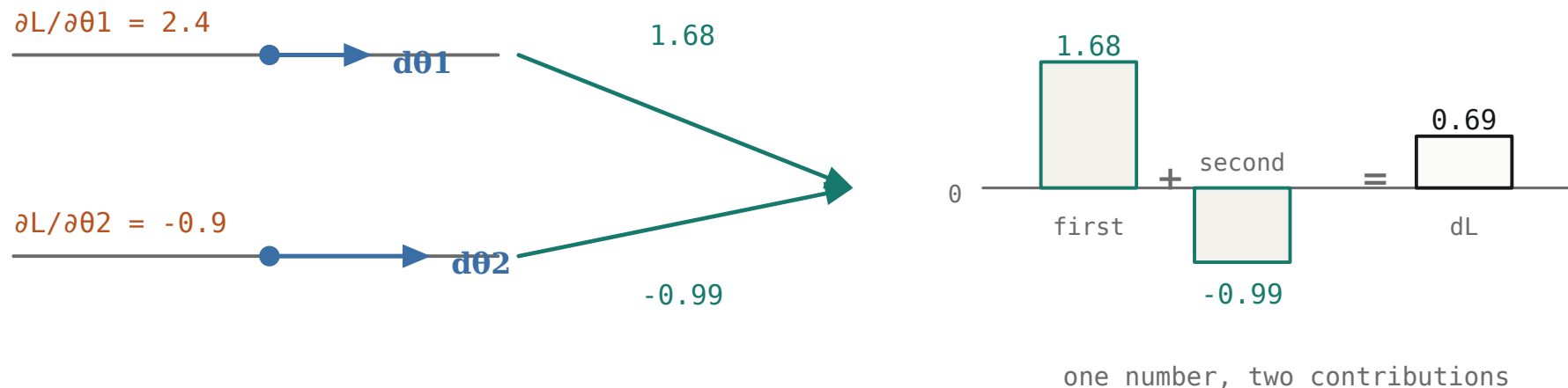
changed?

Fitting a **fixed** dataset: which quantity does the update directly adjust?

- (A) the target y^*
- (B) the model weight θ
- (C) the observed input x

From Derivatives to Learning

A change has several contributions



$$dL = \frac{\partial L}{\partial \theta_1} d\theta_1 + \frac{\partial L}{\partial \theta_2} d\theta_2$$

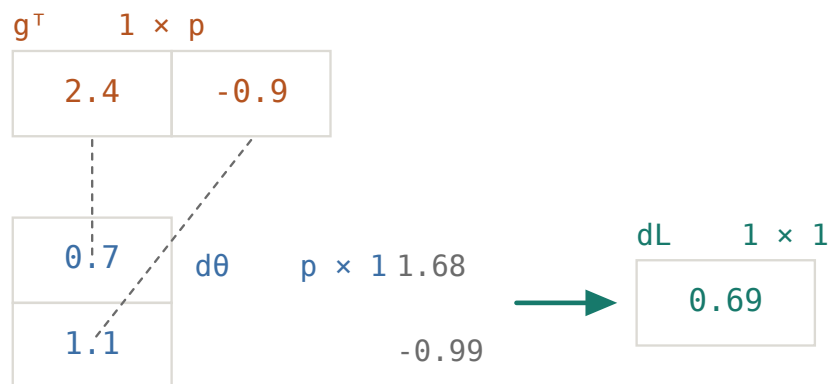
$\partial L / \partial \theta_j$ how sensitive L is to θ_j , at the current θ

$d\theta_j$ the change we make to that one parameter

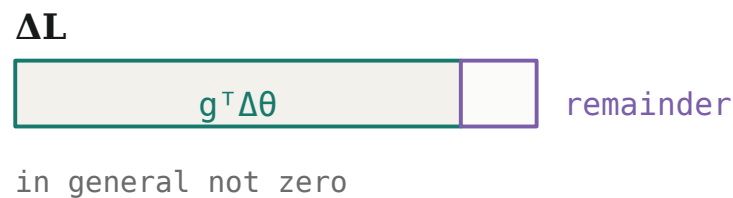
the sum the local linear model, not the exact change

From Derivatives to Learning

The sum is an inner product



the same pairing, for a finite step



$$dL = g^T d\theta, \quad L(\theta + \Delta\theta) - L(\theta) = g^T \Delta\theta + o(\|\Delta\theta\|)$$

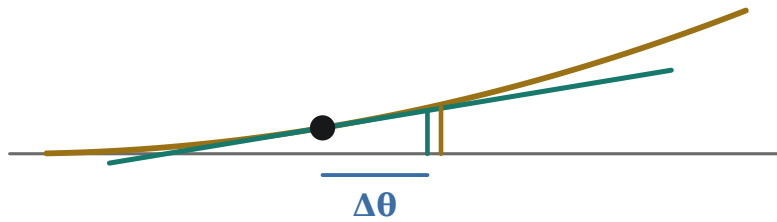
g^T 1 × p, a row of rates

$d\theta$ p × 1, a column of increments

$g^T d\theta$ 1 × 1: the differential, exact $o(\|\Delta\theta\|)$ what a finite step leaves over

From Derivatives to Learning

Derivative or finite change?



tangent
the loss itself

Which expression is an **exact** identity at a differentiable point?

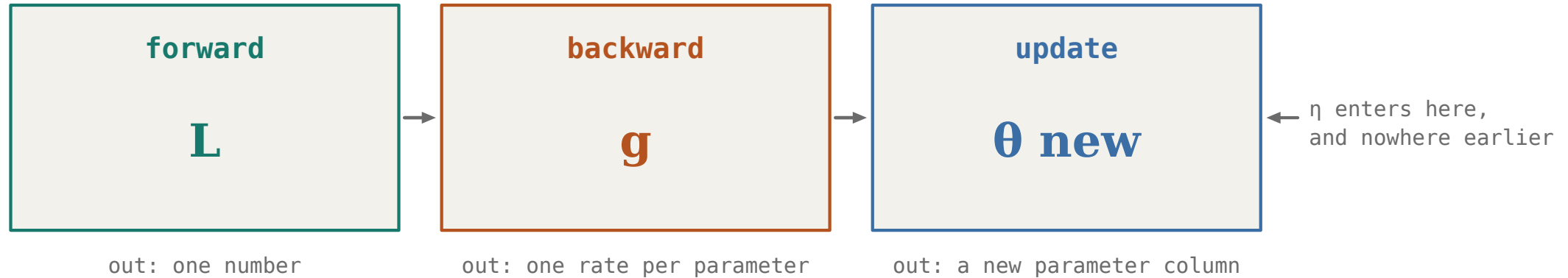
(A) $dL = g^\top d\theta$

(B) $\Delta L = g^\top \Delta\theta$ for every step $\Delta\theta$

(C) $dL = \|g\|$

From Derivatives to Learning

Three jobs in one learning step



(i) evaluate L (ii) compute $g = \nabla_{\theta} L$ (iii) choose $\Delta\theta$, $\theta_{\text{new}} = \theta + \Delta\theta$

(i) one number out

(ii) one rate per parameter out, and nothing is moved yet

(iii) a new parameter column; this is where η enters

VOTE

From Derivatives to Learning

Where does updating happen?



In ordinary gradient descent, which operation changes a weight?

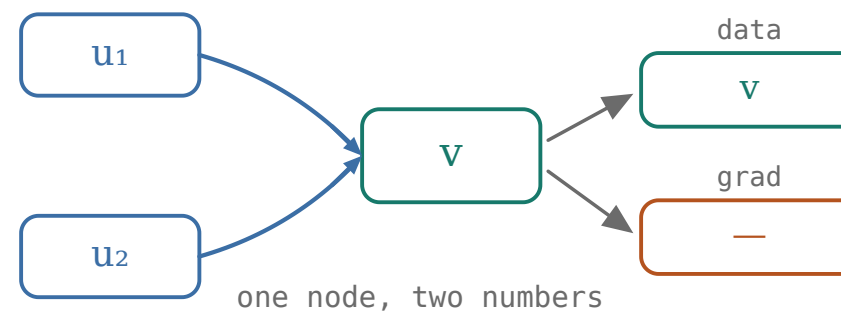
- (A) the forward pass
- (B) the backward pass
- (C) $\theta \leftarrow \theta - \eta g$

Automatic Differentiation

Remembering a computation

one node of the graph, as a record	
data	$v = f(u_1, u_2)$
grad	— nothing yet
parents	u_1, u_2
backward	$\partial v / \partial u_1, \partial v / \partial u_2$

the same record, in place

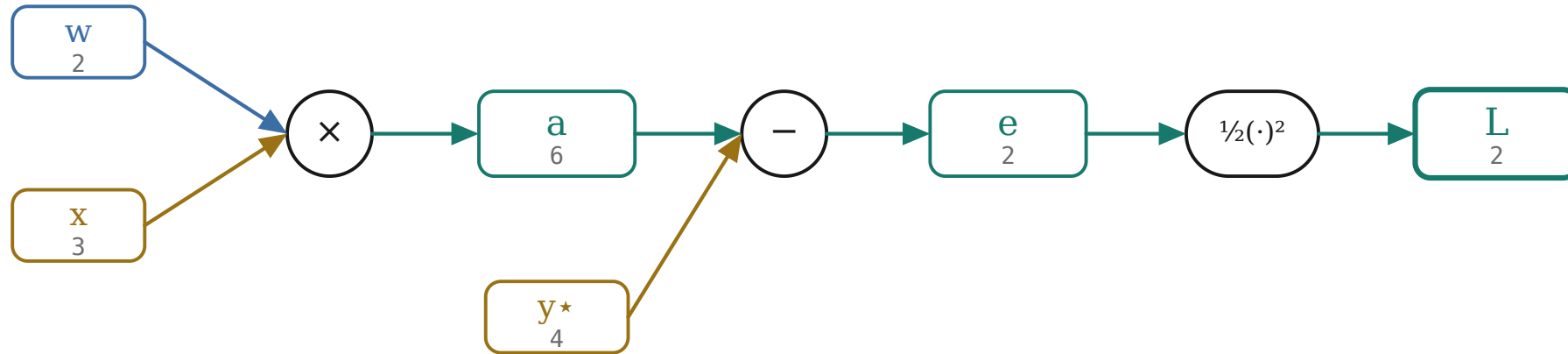


$$v = f(u_1, \dots, u_m), \quad \bar{v} = \frac{\partial L}{\partial v}$$

v the forward number it holds $u_1, \dots, u_m$ its parents, saved
 $\bar{v}$ a second slot, filled backward $\partial v / \partial u_i$ one local rule per parent

Automatic Differentiation

One graph, forward values



forward values only; nothing differentiated yet

$$a = wx, \quad e = a - y^*, \quad L = \frac{1}{2}e^2$$

w the parameter, here 2 x, y^* given: 3 and 4

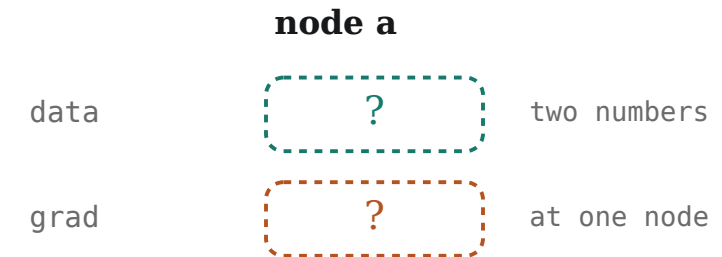
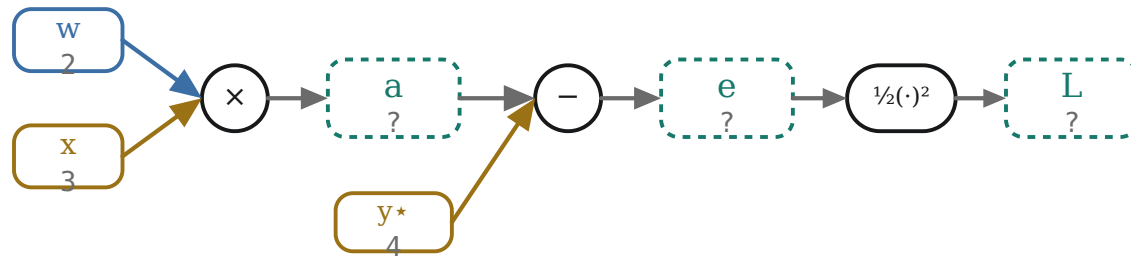
a, e stored forward: 6 and 2 L one scalar out: 2

THE FIXTURE, FORWARD

$w = 2, x = 3, y^* = 4: a = 6, e = 2, L = 2.$

Automatic Differentiation

Value and sensitivity



Same graph, at node a . What is the pair $(a, \bar{a})$?

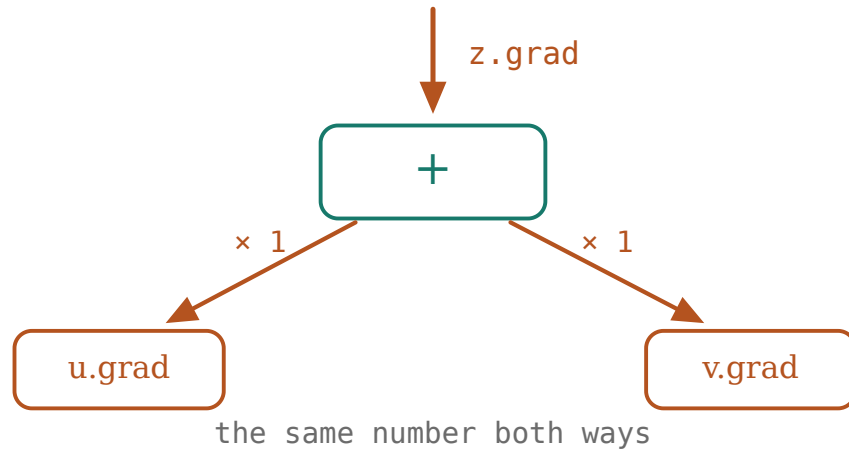
(A) (6, 2)

(B) (2, 6)

(C) (6, 6)

Automatic Differentiation

Local rules become backward callbacks



the local rule, as code

```
def _backward():  
    u.grad += 1.0 * z.grad  
    v.grad += 1.0 * z.grad
```

the local rule

the upstream factor

$$z = u + v : \quad \bar{u} += \bar{z}, \quad \bar{v} += \bar{z}$$

$\bar{z}$ the upstream factor

$\bar{u}, \bar{v}$ the two parent slots

$\partial z / \partial u$ addition's local rule: 1

`+=` add in; a parent has other uses

callback a function *called during* the backward pass

Automatic Differentiation

Multiplication uses saved values

$$w.\text{grad} = a.\text{grad} \cdot x$$

6

w
2

$$\times x = 3$$

2

$$\times 1$$

2

$$\times e = 2$$

1

grad

a
6

e
2

L
2

$$\times w = 2$$

4

x
3

$$x.\text{grad} = a.\text{grad} \cdot w$$

$$z = uv : \quad \bar{u} += v \bar{z}, \quad \bar{v} += u \bar{z}$$

u, v both saved going forward

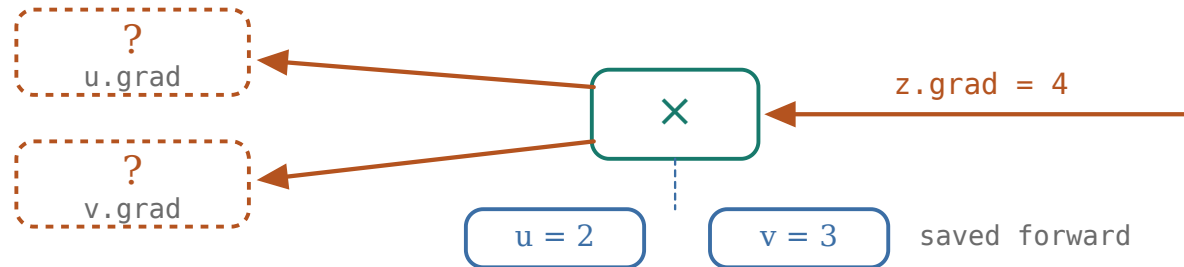
$\bar{z}$ the upstream factor

$v \bar{z}$ into $\bar{u}$: the *other* input

why saved each side needs the other

Automatic Differentiation

A missing upstream factor



$z = uv$ with $u = 2$, $v = 3$, and $\bar{z} = 4$ arriving. What lands in $\bar{u}$?

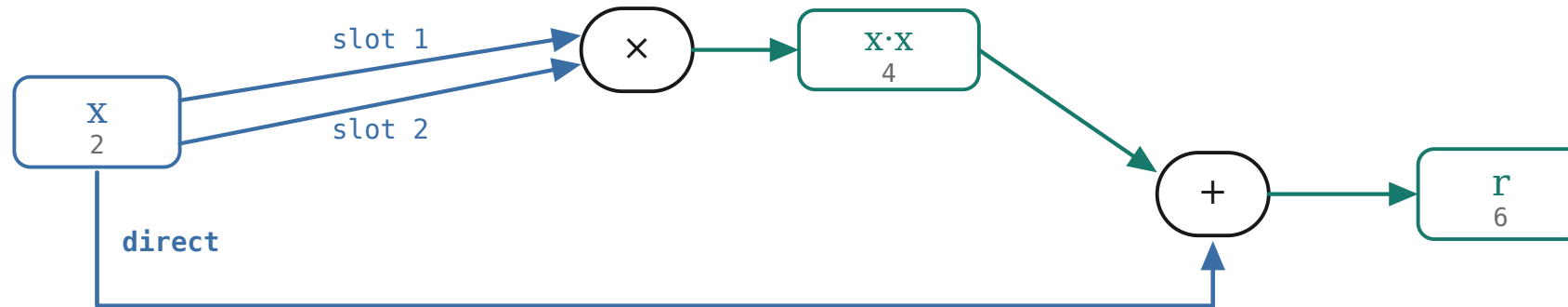
(A) 3

(B) 8

(C) 12

Automatic Differentiation

One variable, several uses



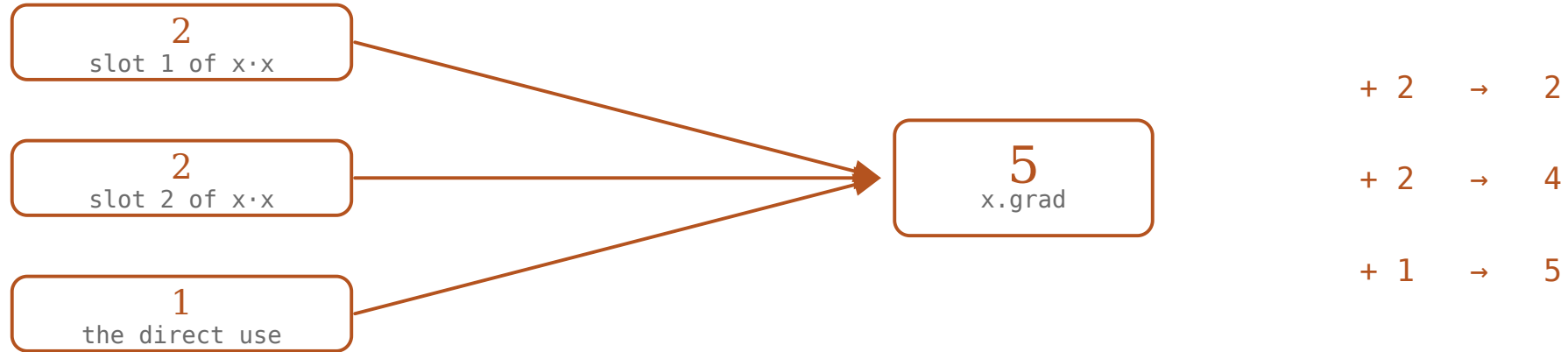
three edges out of x , visited once in traversal

$$r = x \cdot x + x$$

x one node, used three times $x \cdot x$ two argument slots
 $+x$ a third, direct use r at $x = 2$ it stores 6

Automatic Differentiation

Contributions accumulate



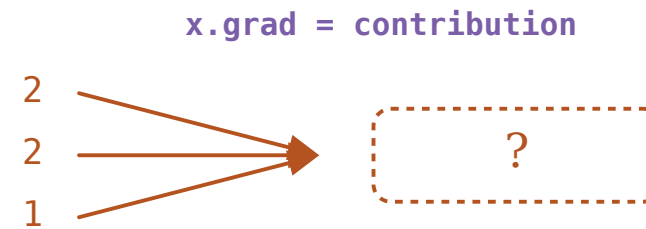
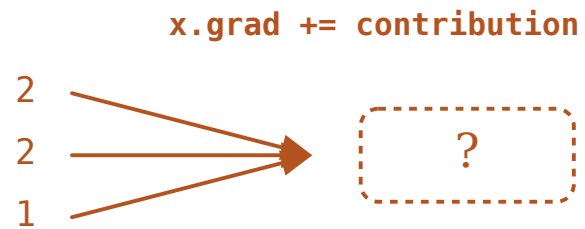
every use reports; the slot adds, never replaces

$$\frac{dr}{dx} = \underbrace{x}_{\text{slot 1}} + \underbrace{x}_{\text{slot 2}} + \underbrace{1}_{\text{direct}} = 5 \quad (x = 2)$$

x the node, used three times $x \cdot x$ two slots, each giving x
 $+x$ the direct use, giving 1 dr/dx the sum, at $x = 2$: 5

Automatic Differentiation

Assignment or addition?



For $r = x \cdot x + x$ at $x = 2$, what is dr/dx ?

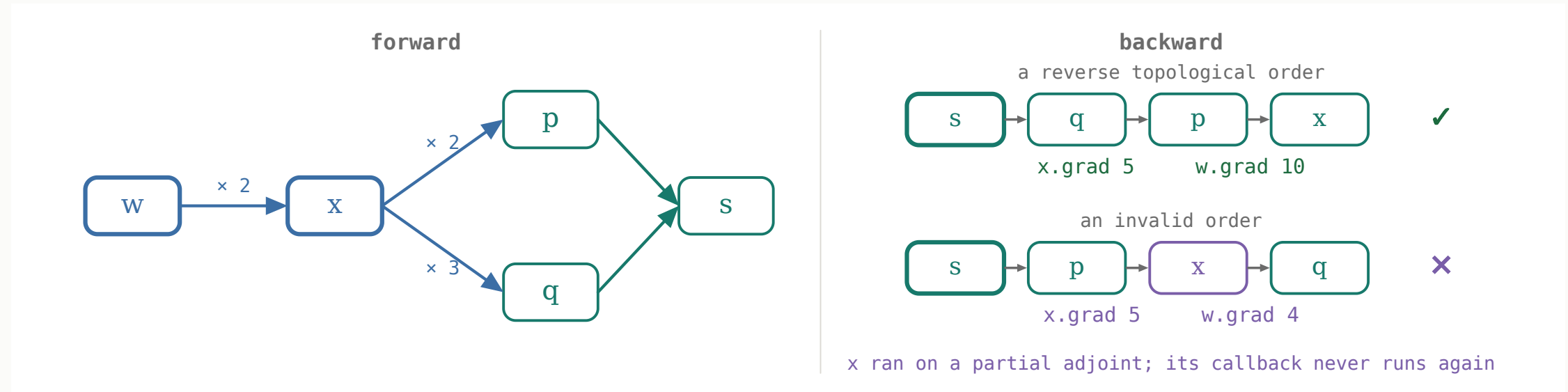
(A) 1

(B) 3

(C) 5

Automatic Differentiation

Reverse topological order



$$\bar{u} = \sum_{s: u \rightarrow v} \bar{v} \frac{\partial v}{\partial s}$$

$s: u \rightarrow v$ one input slot; vary it with the others fixed

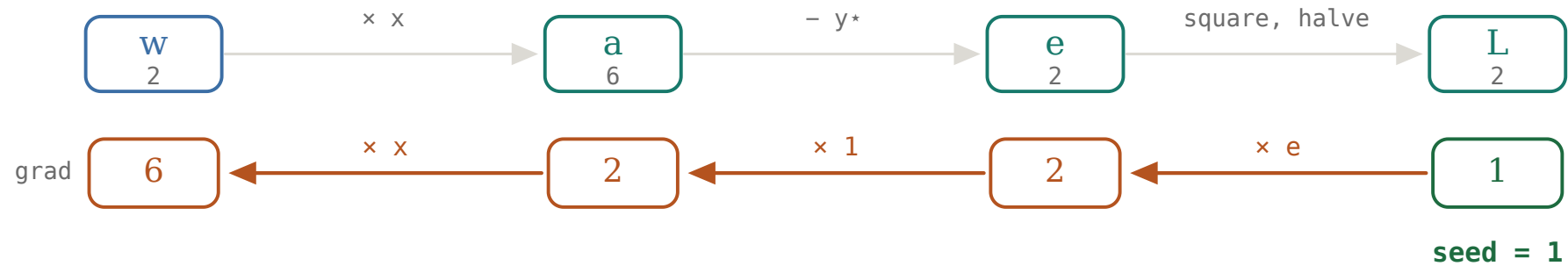
$\bar{v}$ must be final already

the sum complete when all uses report

order any reverse topological one

Automatic Differentiation

The loss starts with one



the only number we set by hand is the seed

$$\bar{L} = \frac{\partial L}{\partial L} = 1$$

$\bar{L}$ the seed: the one adjoint we set by hand

1 a derivative, not the value of L

the rest start at 0, filled by a local rule

Automatic Differentiation

Trace the backward schedule



- | | | | | |
|---|-------------------------------------|---|----------------------------------|---|
| 0 | zero every slot, seed $\bar{L} = 1$ | → | writes nothing | |
| 1 | L's callback | → | writes $\bar{e} = 2$ | a seed runs nothing: 1 into $\bar{L}$ moves no slot |
| 2 | e's callback | → | writes $\bar{a} = 2$ | each callback writes one slot upstream |
| 3 | a's callback | → | writes $\bar{w} = 3 \cdot 2 = 6$ | |

$\bar{L} \leftarrow 1,$ then run L, e, a

seed a number put in a slot; it runs nothing

callback what writes into the slot below it

One step, one slot. $\bar{w} = 6$ at the end.

Automatic Differentiation

Now two runs that break a rule



which run survives the rules?

Zero the slots, then seed and run. Which gives the right $\bar{w}$?

(A) $\bar{L} = L: L, e, a$

(B) $\bar{L} = 1: L, e, a$

(C) $\bar{L} = 1: L, a, e$

Automatic Differentiation

ReLU reuses its forward state

$$a = \max(0, z)$$



no ordinary derivative at 0
the engine returns 0 by convention

saved z	mask	what leaves
2	1	a.grad
-2	0	0

the mask reads z, never the sign of a.grad

$$a = \max(0, z), \quad \bar{z} = \bar{a} \cdot \mathbf{1}_{z>0}$$

z the saved preactivation

$\mathbf{1}_{z>0}$ 1 if $z > 0$, else 0

$\bar{a}$ whatever arrives, any sign

at $z = 0$ no derivative; 0 by convention

Automatic Differentiation

An open gate can pass a negative gradient



$a = \max(0, z)$ with $z = 2$ saved, and $\bar{a} = -3$ arrives. What is $\bar{z}$?

(A) 0

(B) -3

(C) 3

Automatic Differentiation

Checking a computed derivative



$$\frac{L(w + \tau) - L(w - \tau)}{2\tau} \quad \text{vs} \quad \frac{L(w + \tau) - L(w)}{\tau}$$

τ the probe step, 1/100
centred exact here
one-sided off by $\tau x^2/2$
 w restored afterwards

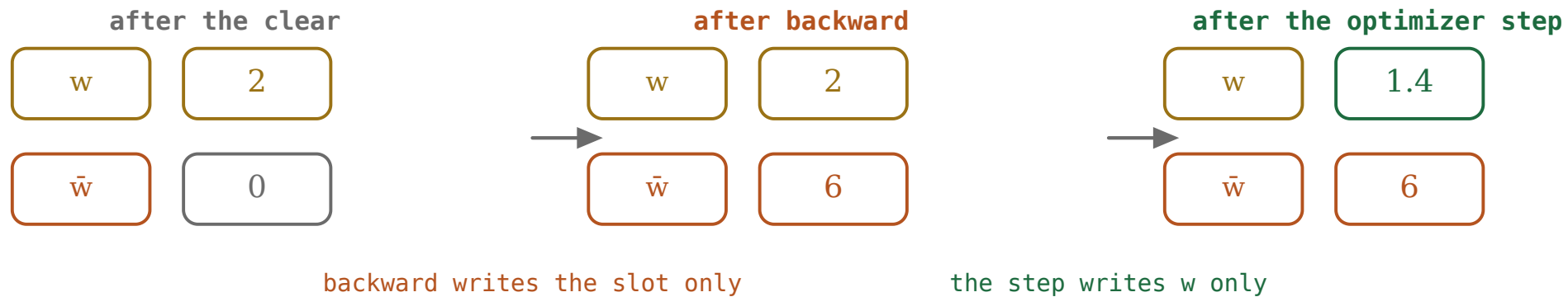
ONE DERIVATIVE, THREE ESTIMATES

By hand $\bar{a} x = 6$; the engine's grad, 6; the centred probe, 6.

One-sided: 6.045, over by $\tau x^2/2 = 0.045$. Agreement is evidence only.

Automatic Differentiation

Backward fills slots; plain SGD moves w



plain SGD: $w \leftarrow 2 - 0.1 \cdot 6 = 1.4$ the slot still reads 6
but 6 is the gradient AT $w = 2$. Nobody has computed the one at 1.4 yet.

backward: $\bar{w} += \dots$,

plain SGD: $w \leftarrow w - \eta \bar{w}$

w the parameter; only the optimizer assigns to it

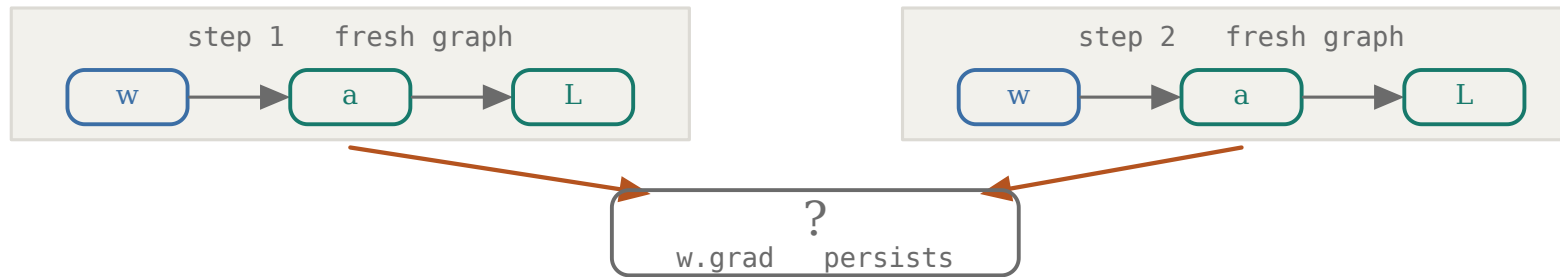
$\bar{w}$ its slot; only backward and the clear write it

η step size, here 0.1; chosen, not computed

$\bar{w} = 6$ read at $w = 2$; it is not the gradient at 1.4

Automatic Differentiation

Why reset gradients?



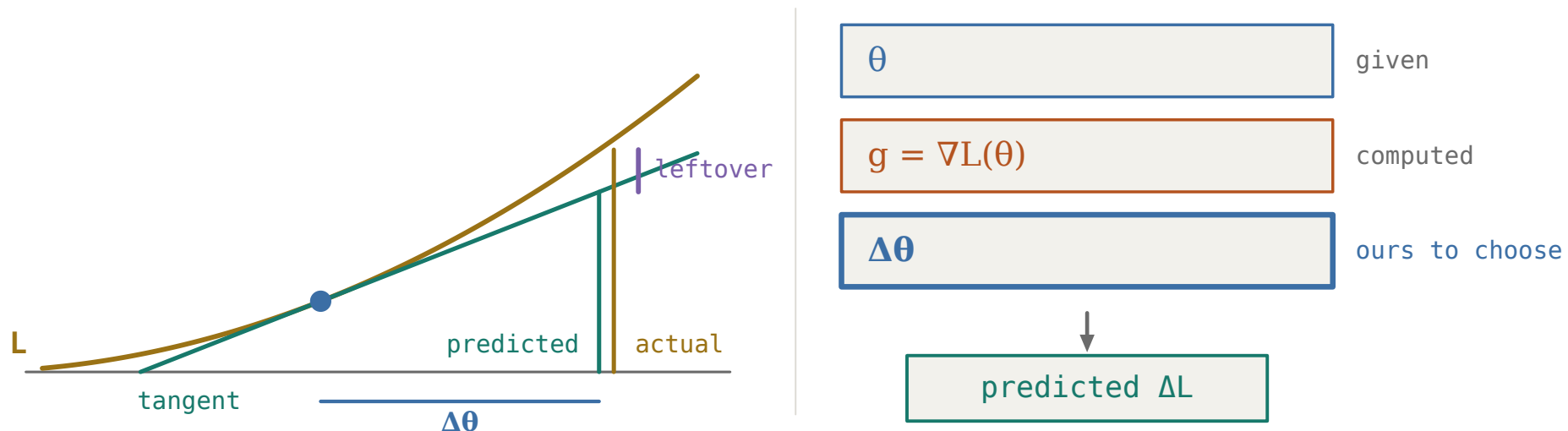
Two fresh graphs at unchanged w , no clearing. What is in $\bar{w}$?

- ☐ (A) this step's gradient only
- ☐ (B) the old and the new, added
- ☐ (C) w updates itself

RECALL

Gradient Descent and Step Size

Choosing a parameter change



$$\Delta L = L(\theta + \Delta\theta) - L(\theta) \approx g^T \Delta\theta$$

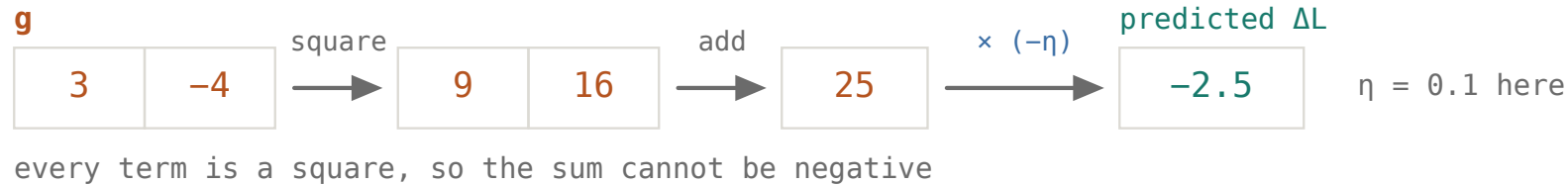
$\Delta\theta$ $p \times 1$: the step, and the only thing we choose

g $p \times 1$: $\nabla_{\theta} L$ at the current θ , fixed while we choose

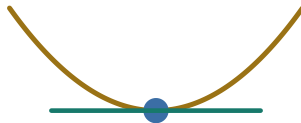
$\approx$ the linear part only; the leftover is dropped, not zero

Gradient Descent and Step Size

Why the negative gradient?

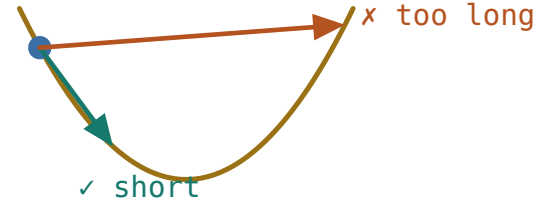


if $g = 0$



prediction θ , and
 $-\eta g$ moves nothing

if the step is long



the model said down;
the curve went up

$$\Delta\theta = -\eta g \implies g^T \Delta\theta = -\eta g^T g = -\eta \sum_{j=1}^p g_j^2 < 0$$

$\eta > 0$ our step size, never folded into g

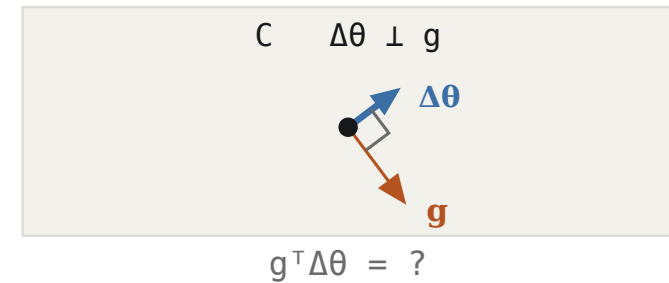
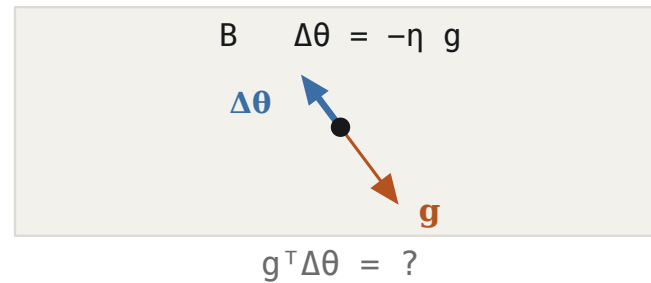
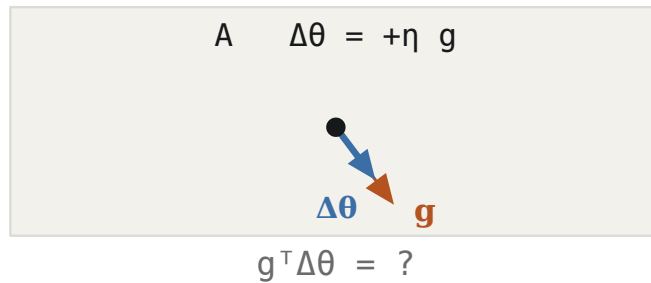
$\sum_j g_j^2$ a sum of squares, so it is ≥ 0

< 0 needs only $g \neq 0$ and $\eta > 0$

$\Delta L < 0$ a different claim: the loss itself also needs a short step

Gradient Descent and Step Size

Which direction decreases the local model?



For $g \neq 0$, $\eta > 0$: which step makes the predicted $g^T \Delta\theta$ negative?

- ☐ A $\Delta\theta = +\eta g$
- ☐ B $\Delta\theta = -\eta g$
- ☐ C a nonzero step orthogonal to g

Gradient Descent and Step Size

The gradient-descent update



$$1 - 0.1 \times 3 = 0.7$$

$$2 - 0.1 \times (-4) = 2.4$$

one entry at a time, with the same η

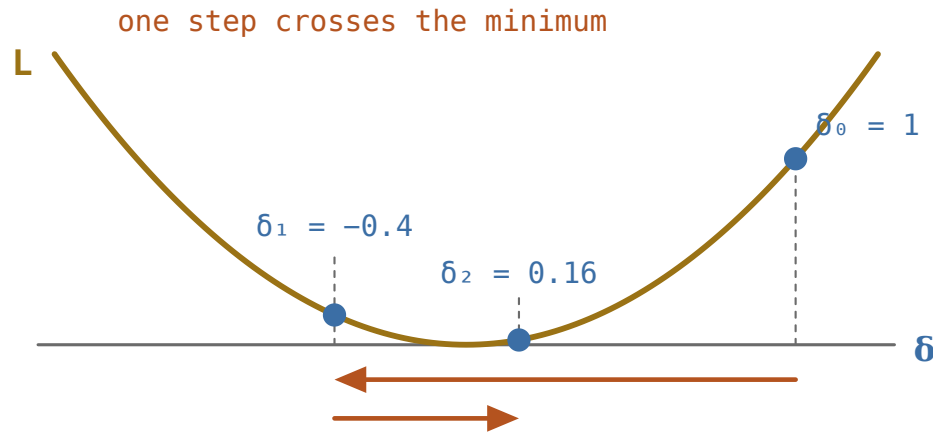
$$\theta_{t+1} = \theta_t - \eta_t g_t, \quad g_t = \nabla_{\theta} L(\theta_t)$$

θ_t the current column, $p \times 1$ g_t read at θ_t , not at θ_{t+1}

θ_{t+1} the new column, same shape η_t one positive number per step

Gradient Descent and Step Size

A scalar quadratic exposes overshoot



$$\lambda = 4, \quad \eta = 0.35$$

$$g = \lambda \delta$$

$$\begin{aligned} \delta - \eta g &= \delta - \eta \lambda \delta \\ &= (1 - \eta \lambda) \delta \end{aligned}$$

$$\eta \lambda = 1.4 > 1 \quad 1 - \eta \lambda = -0.4$$

the step is longer than the distance to the minimum

$$L = \frac{1}{2} \lambda \delta^2, \quad g = \lambda \delta, \quad \delta_{t+1} = \delta_t - \eta \lambda \delta_t = (1 - \eta \lambda) \delta_t$$

δ_t $\theta_t - \theta^*$, one number here

$\lambda > 0$ the curvature of this quadratic

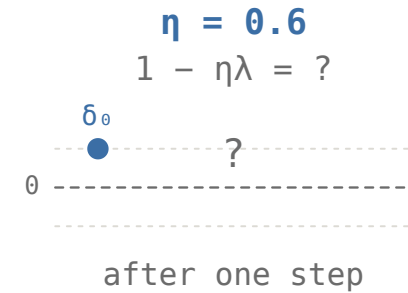
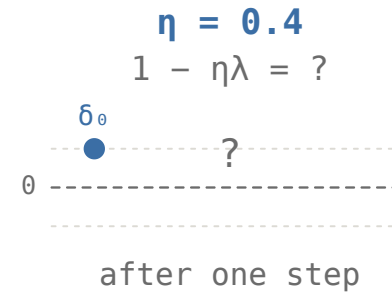
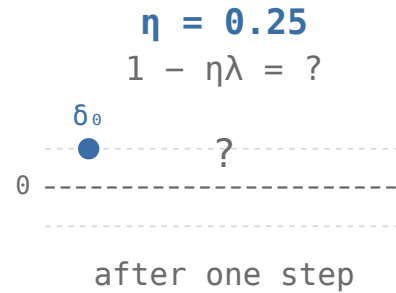
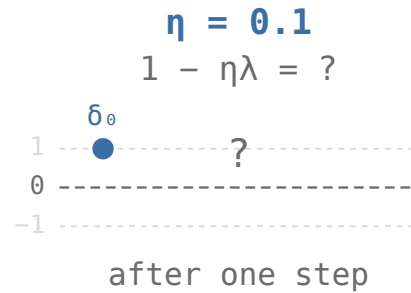
$1 - \eta \lambda$ the multiplier: it decides every later step

WORKED EXAMPLE

$\eta \lambda = 0.35 \times 4 = 1.4 > 1$, so $1 - \eta \lambda = -0.4$.

Gradient Descent and Step Size

Four step-size regimes



Take $\lambda = 4$. Which pair is (converges while oscillating, diverges)?

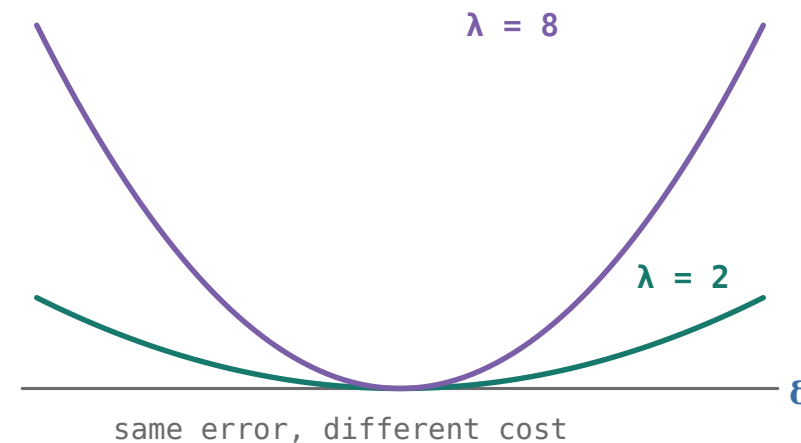
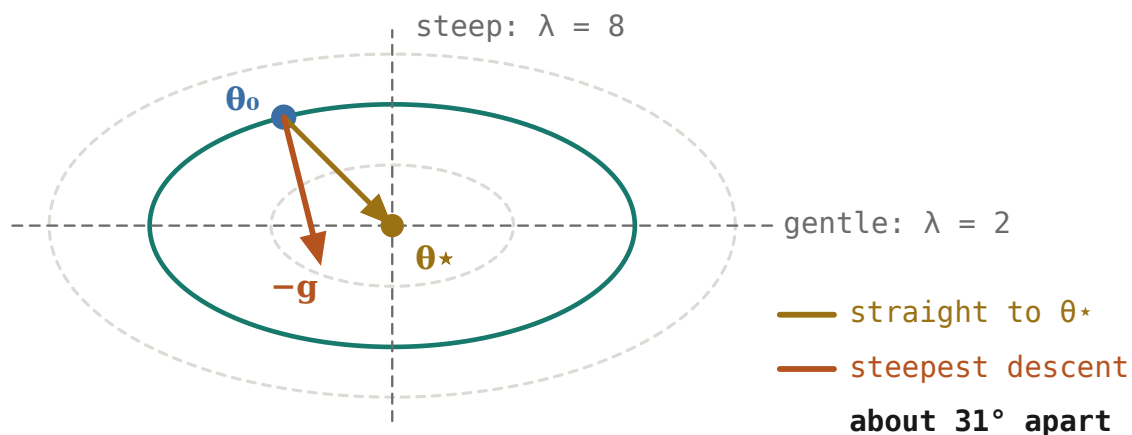
(A) (0.4, 0.6)

(B) (0.1, 0.25)

(C) (0.6, 0.4)

Gradient Descent and Step Size

Several curvature directions



$$L = \frac{1}{2} \delta^T H \delta, \quad g = H \delta, \quad H = Q \Lambda Q^T$$

H symmetric, positive definite,
 $p \times p$

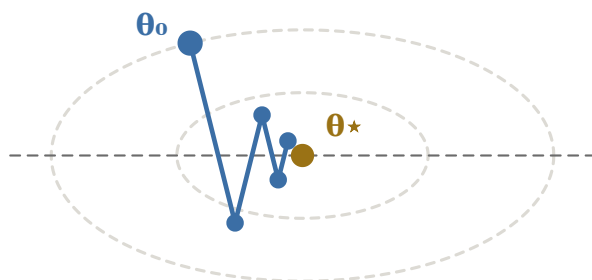
Q orthonormal eigenvectors q_i

Λ $\text{diag}(\lambda_1, \dots, \lambda_p)$, each $\lambda_i > 0$

$g = H \delta$ for $\delta \neq 0$, parallel to δ exactly
when it is an eigenvector

Gradient Descent and Step Size

Each eigenmode has one multiplier



$\eta = 0.2$, four steps

mode q_1 , $\lambda = 2$

factor 0.6

start

0

same sign every step

mode q_2 , $\lambda = 8$

factor -0.6

start

0

sign flips every step

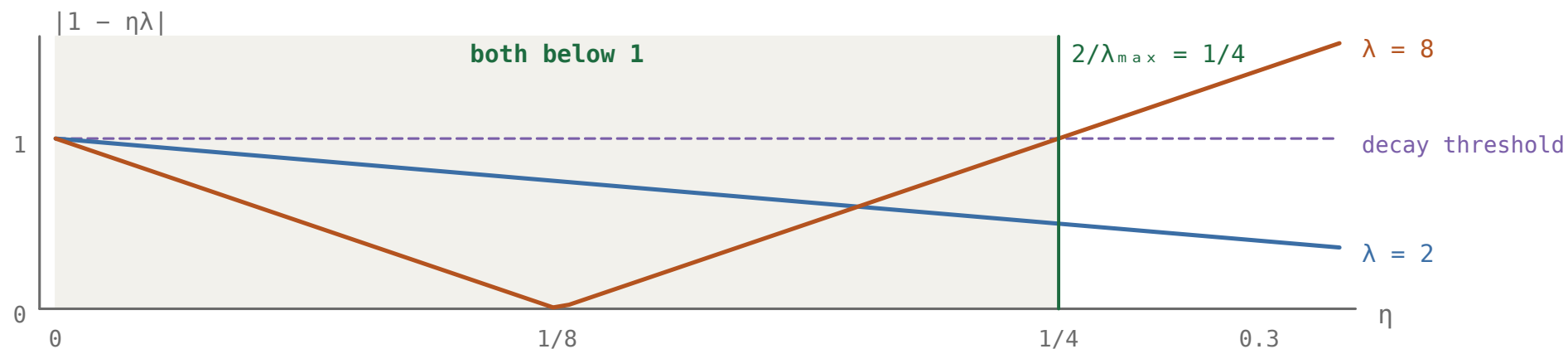
$$\delta_{t+1} = (I - \eta H) \delta_t, \quad \delta_t = \sum_i c_i^{(t)} q_i \Rightarrow c_i^{(t+1)} = (1 - \eta \lambda_i) c_i^{(t)}$$

$c_i^{(t)}$ the error's coordinate along q_i $1 - \eta \lambda_i$ one factor per mode; modes never mix

If some $\lambda_i = 0$, that factor is 1: the mode keeps its error for every η .

Gradient Descent and Step Size

When does every mode decay?



the steep mode $\lambda = 8$ leaves the band first, so it sets the limit
all of this is about one fixed $H > 0$, and nothing else

$$|1 - \eta\lambda_i| < 1 \text{ for every } i \iff 0 < \eta < \frac{2}{\lambda_{\max}}$$

λ_i the curvatures of one fixed
 $H \succ 0$

$\lambda_{\max}$ the largest; the steep mode
binds first

scope this quadratic only, not any
loss, not a network

Gradient Descent and Step Size

Which curvature limits the step?

H		
2	0	$\lambda = 2$
0	8	$\lambda = 8$

which step sizes are safe?

_____ η

0

$H = \text{diag}(2, 8)$. Which η sends the error to zero from every start?

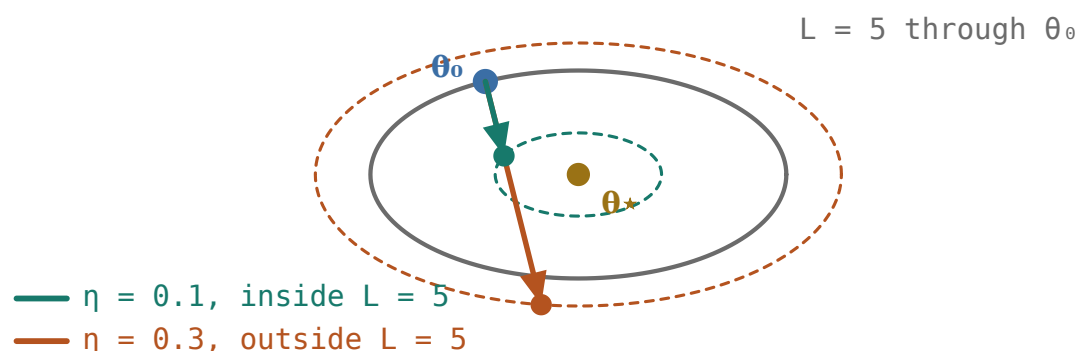
(A) $0 < \eta < 1$

(B) $0 < \eta < \frac{1}{4}$

(C) $0 < \eta \leq \frac{1}{4}$

Gradient Descent and Step Size

A complete two-parameter step



start: $\theta_0 = (0, 0)$, $L = 5$

η	θ_1	$L(\theta_1)$	
0.1	$(0.2, -0.8)$	0.8	falls
0.3	$(0.6, -2.4)$	8	rises

same start, same gradient, one number changed

$$g_0 = H(\theta_0 - \theta^*) = (-2, 8), \quad \theta_1 = \theta_0 - \eta g_0$$

$\theta^* = (1, -1)$ the minimiser; $H = \text{diag}(2, 8)$ $\theta_0 = 0$ so $\delta_0 = (-1, 1)$

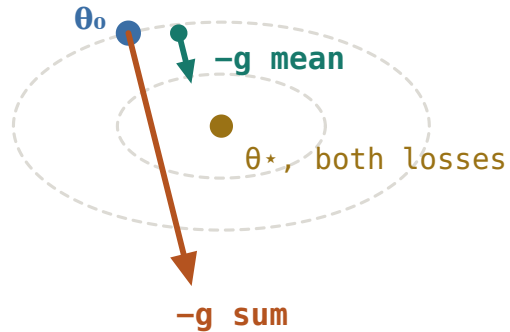
g_0 a column: 8 steep, -2 gentle

WORKED EXAMPLE

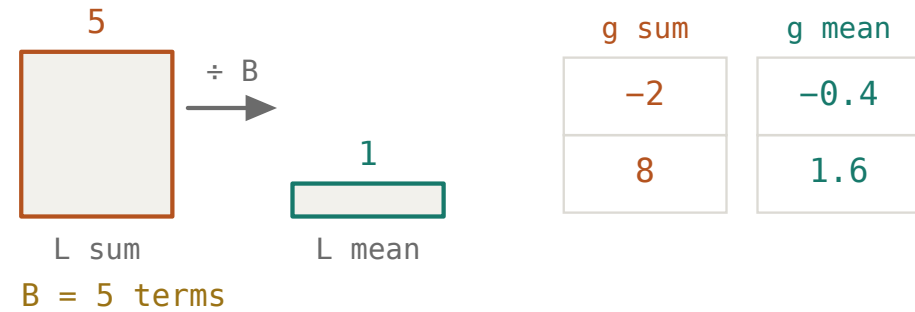
$\eta = 0.1$ takes L from 5 to 0.8. $\eta = 0.3$ raises it to 8: that is **above** the bound, steep factor -1.4 , so repeating it diverges.

Gradient Descent and Step Size

Scaling the loss scales the update



same direction, one fifth the length



dividing by five keeps every comparison, so the minimiser is the same point for both losses

$$L_{\text{mean}} = \frac{1}{B} L_{\text{sum}} \implies g_{\text{mean}} = \frac{1}{B} g_{\text{sum}}$$

B how many terms the sum has; $B = 5$ here

θ^* unmoved: dividing by a positive B keeps every comparison

g_{mean} every entry $1/B$ as large

every term is scaled, not just one of them

Gradient Descent and Step Size

Which pair reproduces the step?

$$\begin{aligned} \text{subtract: } \eta \text{ sum} \times g \text{ sum} &= 0.1 \times (-2, 8) = (-0.2, 0.8) \\ g \text{ mean} &= (-0.4, 1.6), \quad B = 5 \end{aligned}$$

- | | | |
|---|------------------------------|-----------------------|
| A | $\eta = 0.1, g \text{ mean}$ | subtract $\eta g = ?$ |
| B | $\eta = 0.5, g \text{ mean}$ | subtract $\eta g = ?$ |
| C | $\eta = 0.5, g \text{ sum}$ | subtract $\eta g = ?$ |

which row reproduces it?

Which pair gives the same update $\theta - \eta g$ as the sum loss?

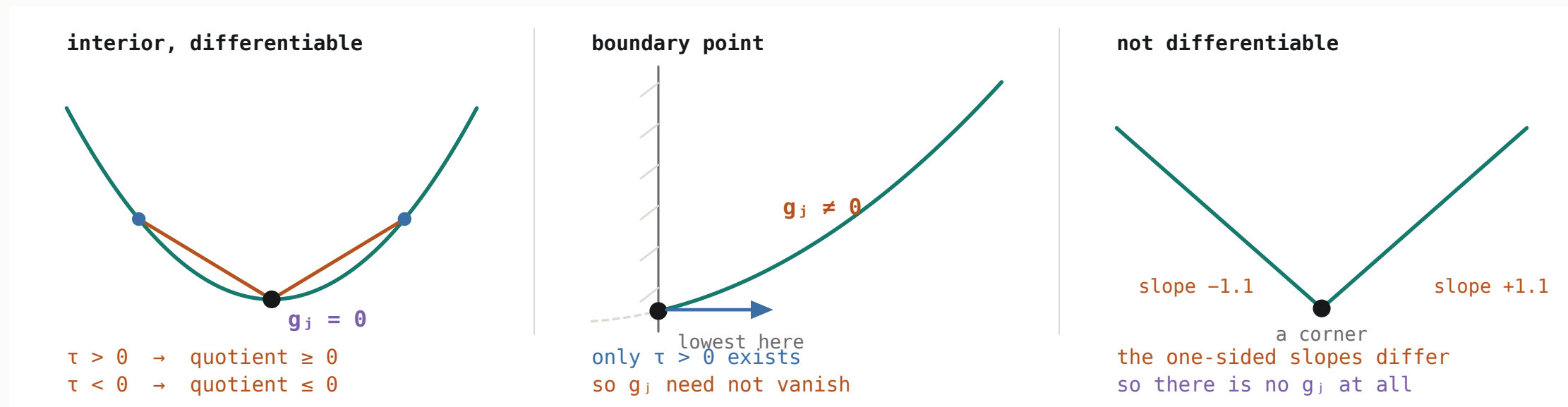
(A) 0.1 with g_{mean}

(B) 0.5 with g_{mean}

(C) 0.5 with g_{sum}

Convexity and Optimality

What zero gradient tells us



interior, L differentiable, local minimum :

$$\frac{L(\theta + \tau e_j) - L(\theta)}{\tau} \begin{cases} \geq 0, & \tau > 0 \\ \leq 0, & \tau < 0 \end{cases} \implies g_j = 0$$

e_j the j -th coordinate direction

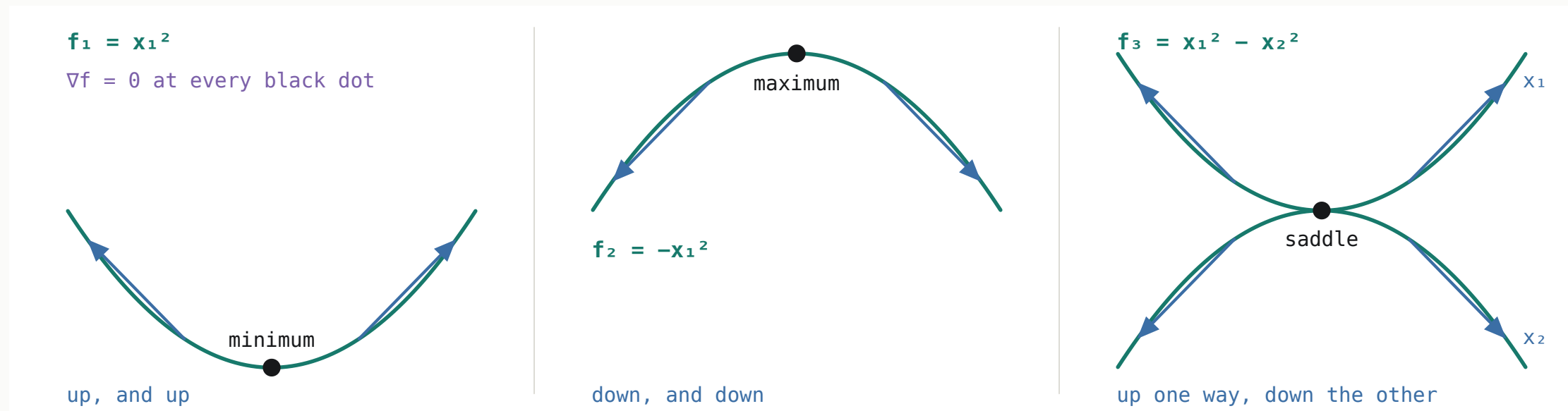
g_j one limit, so ≥ 0 and ≤ 0 at once

τ a sufficiently small nonzero probe step, either sign

a corner no limit, so no g_j to set to zero

Convexity and Optimality

Stationary points have different roles



$$f_1 = x_1^2, \quad f_2 = -x_1^2, \quad f_3 = x_1^2 - x_2^2, \quad \nabla f_k(0) = 0$$

f_1 up along $+x_1$ and along $-x_1$: a minimum

f_3 up along x_1 , down along x_2 : a saddle

f_2 down along both: a maximum

$\nabla f_k(0)$ the same zero in all three, so it cannot tell them apart

Convexity and Optimality

Is a PSD Hessian at one point enough?

For $f(x) = -x^4$ at 0: $f'(0) = f''(0) = 0$, so $0 \succeq 0$ passes. Is 0

- (A) a local minimum
- (B) a local maximum
- (C) neither

three readings at $x = 0$, the same for both

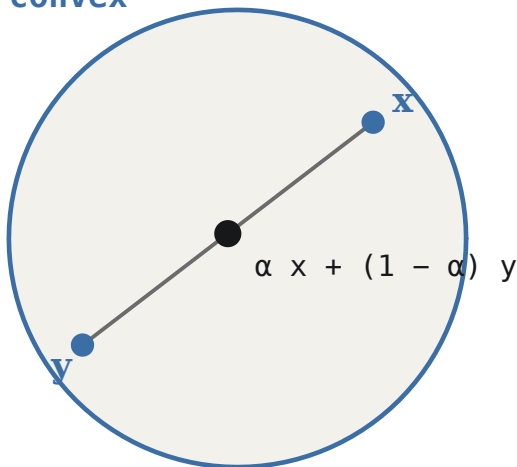
	$f(x) = -x^4$	$g(x) = x^4$
value	0	0
slope	0	0
curvature	0	0

DEFINITION

Convexity and Optimality

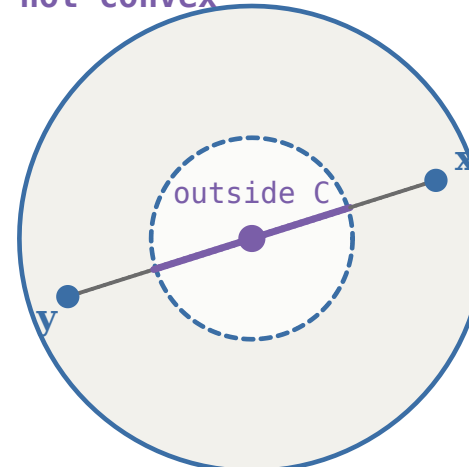
A convex domain contains segments

convex



every segment stays inside

not convex



this one segment leaves it

$$x, y \in C, \quad \alpha \in [0, 1] \quad \implies \quad \alpha x + (1 - \alpha)y \in C$$

C a set of points, the domain a convex function needs

$\alpha x + (1 - \alpha)y$ every point of the segment, not only its two ends

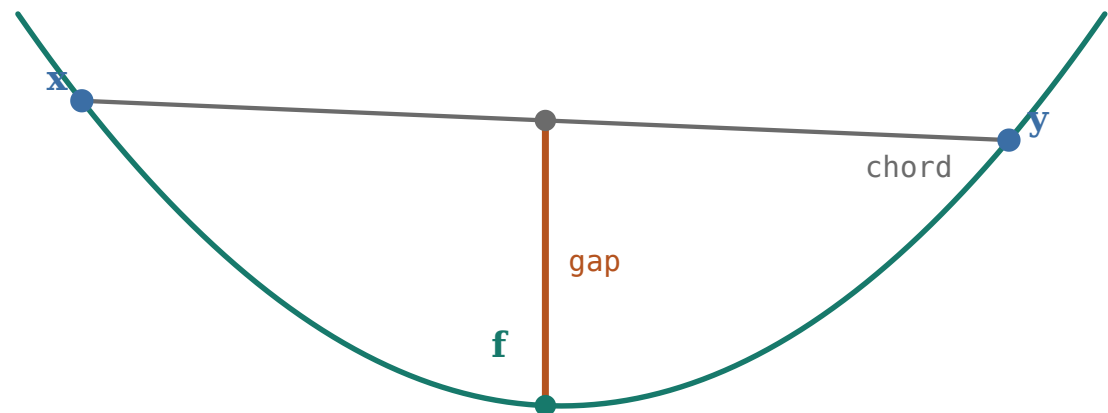
α where on the segment you stand: 0 gives y , 1 gives x

not convex one segment leaving C is enough

DEFINITION

Convexity and Optimality

A convex function lies below its chords



the chord is never below f

a domain in two pieces



midpoint: not in the domain

so there is nothing to compare

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y) \quad \text{for all } x, y \in C, \alpha \in [0, 1]$$

left side the function at the interpolated point

the gap chord minus function, never negative

right side the height of the chord there

C convex, or some segment leaves it and the left side has no value

Convexity and Optimality

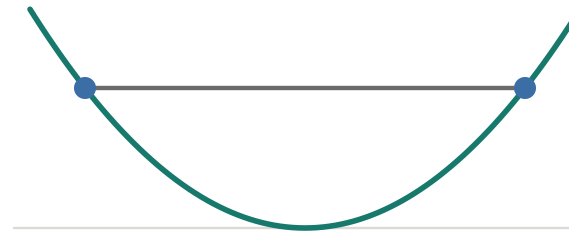
Which picture satisfies the definition?

At $x = -1$ and $y = 1$ with $\alpha = \frac{1}{2}$: which obeys $f(0) \leq \frac{f(-1)+f(1)}{2}$?

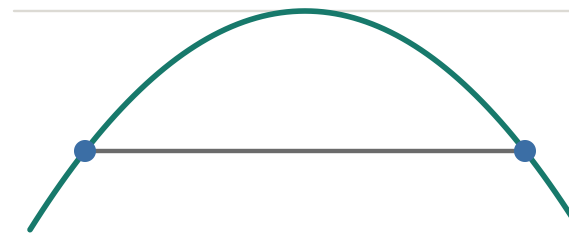
(A) $f = x^2$

(B) $f = -x^2$

(C) both



A: $f = x^2$



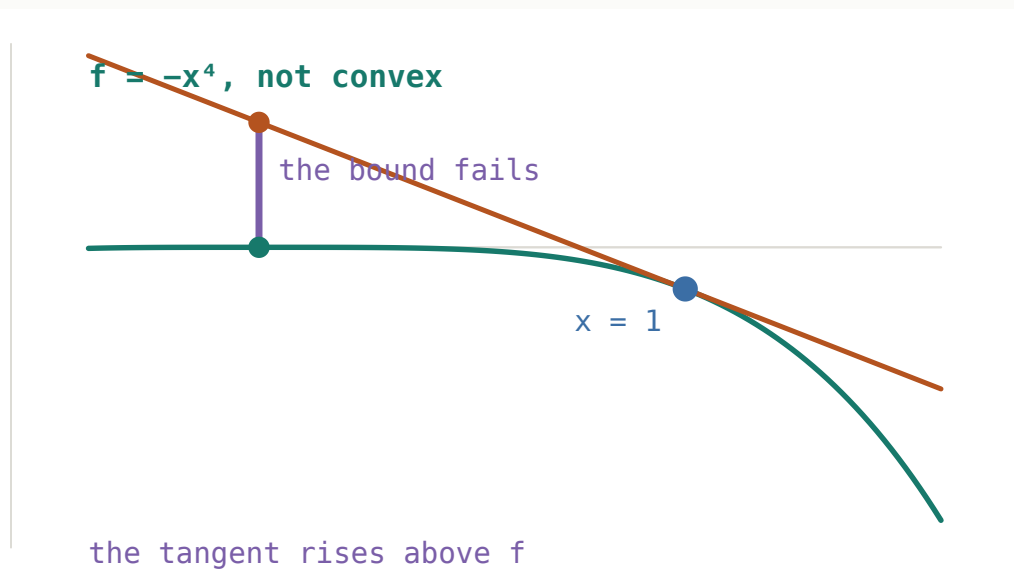
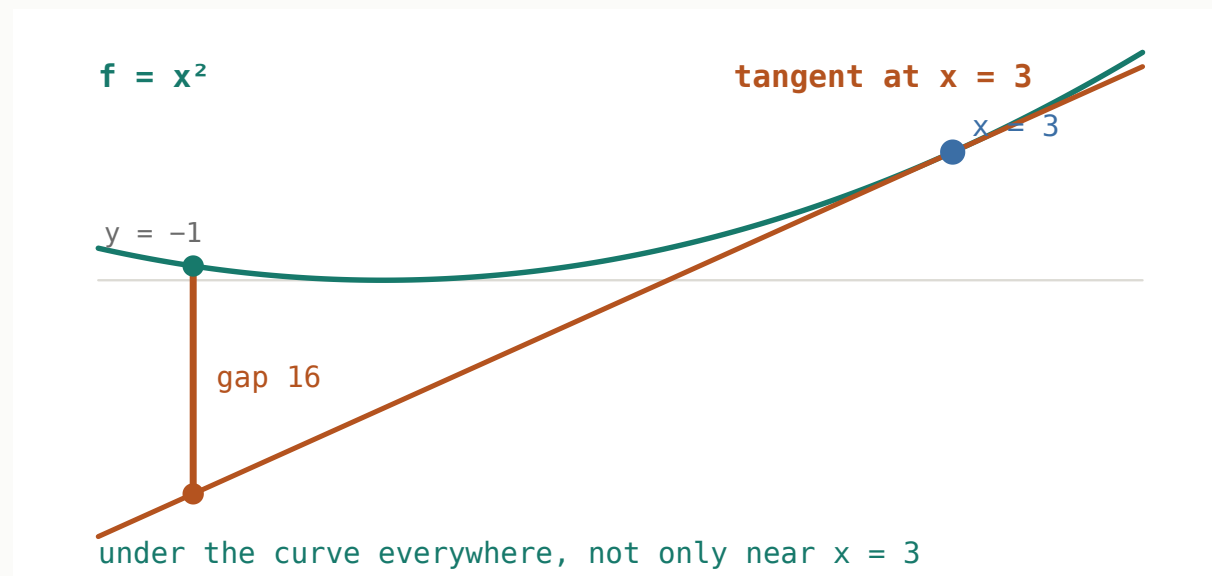
ends at $x = -1$ and $x = 1$

B: $f = -x^2$

KEY INEQUALITY

Convexity and Optimality

A supporting tangent gives a global bound



$$f(y) \geq f(x) + \nabla f(x)^T(y - x) \quad \text{for every } x, y \in C$$

$f(y)$ the value anywhere else in C

$f(x) + \nabla f(x)^T(y - x)$ the tangent's height at y

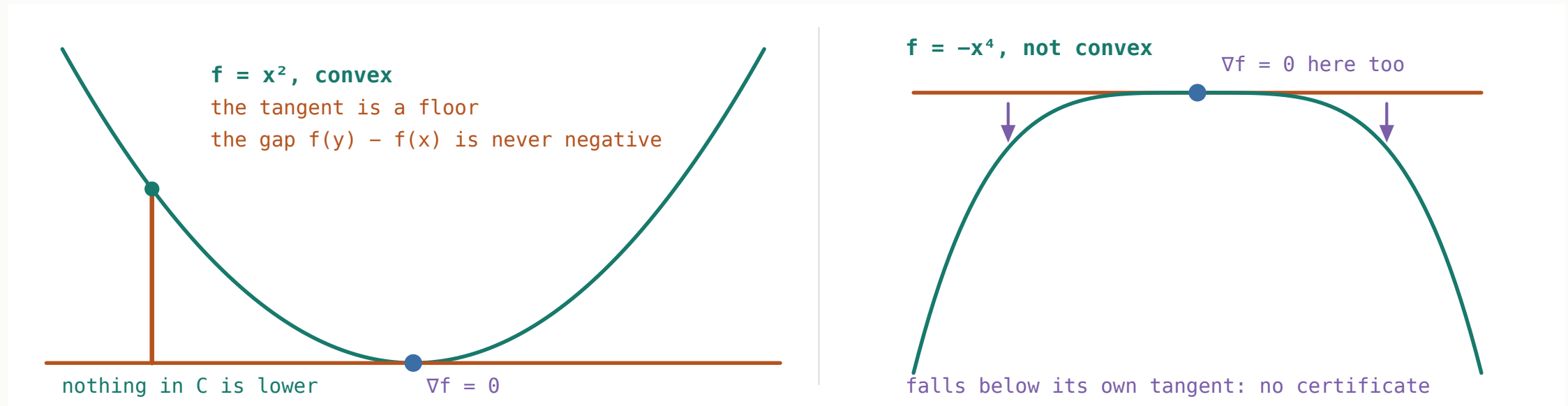
C convex, with f convex and differentiable on it

WORKED EXAMPLE

$f = x^2$ at $x = 3$: the tangent is $9 + 6(y - 3)$. At $y = -1$ it reads -15 , while $f(-1) = 1$. The gap is 16.

Convexity and Optimality

Zero gradient now certifies a minimum



$$\nabla f(x) = 0 \implies f(y) \geq f(x) + 0^T(y - x) = f(x) \quad \text{for every } y \in C$$

$\nabla f(x) = 0$ the hypothesis, put into the tangent bound

$f(y) \geq f(x)$ global: nothing in C is lower

$0^T(y - x)$ the whole linear term, gone

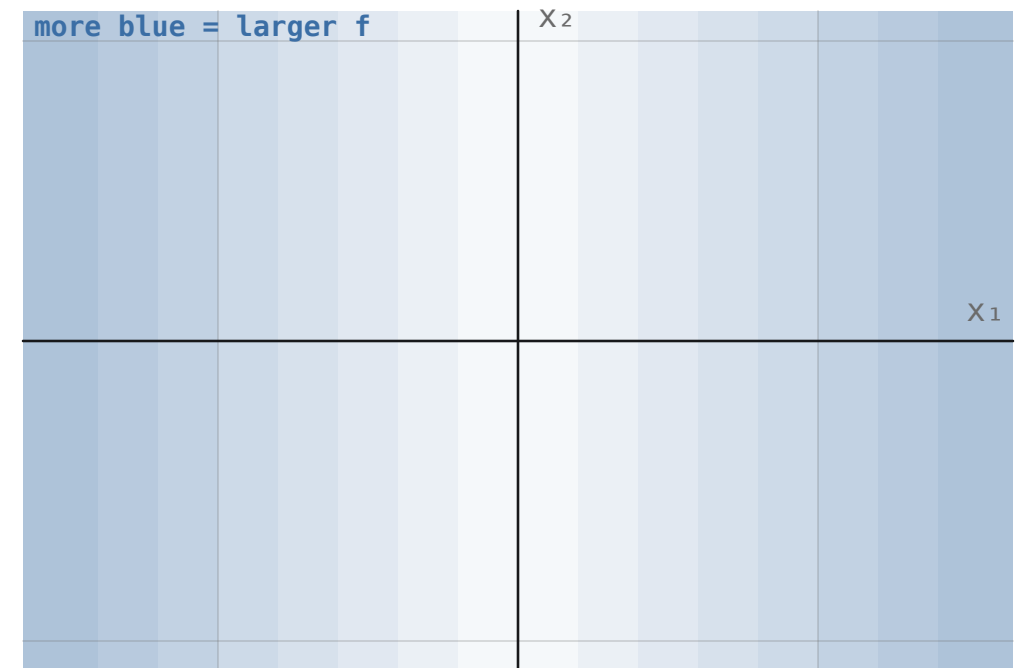
convex the assumption doing the work; without it $\nabla f = 0$ is necessary only, and only in the interior

Convexity and Optimality

Does convex mean a unique minimizer?

$f(x_1, x_2) = x_1^2$ is convex. Which set is exactly its set of minimizers?

- (A) only $(0, 0)$
- (B) every $(0, t)$
- (C) every point

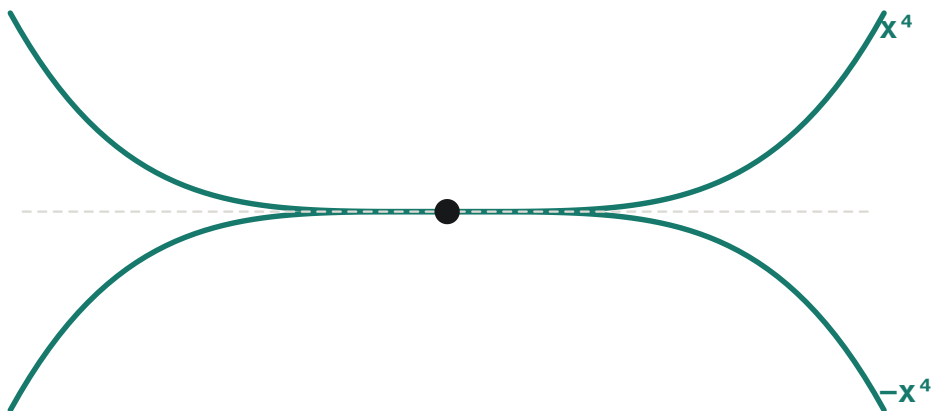


EQUIVALENCE

Convexity and Optimality

Curvature everywhere

flat at the same point



$$\nabla^2(-x^4) = -12x^2$$



negative at every other x : not convex

the dots only sample; the sign is read off the formula

$$\nabla^2(x^4) = 12x^2$$



nonnegative at every x : convex

$f \in C^2$ on an open convex C :

f convex $\iff \nabla^2 f(x) \succeq 0$ for every $x \in C$

$\nabla^2 f(x)$ the Hessian at one point; $\succeq 0$ means PSD there

for every x the quantifier that does the work

C^2 twice continuously differentiable; then it is exact

one point $-x^4$ has $\nabla^2 f(0) = 0$, and $-12x^2 < 0$ everywhere else

Convexity and Optimality

Which conclusion needs which hypothesis

statement	needs	direction
$\nabla L(\theta) = 0$ at an interior local min	interior, differentiable	one way, necessary only
every local min is global	f convex on convex C (no derivative)	one way
$\nabla f(x) = 0 \Rightarrow$ global min on C	f convex AND differentiable on convex C	one way, now sufficient
$\nabla^2 f \geq 0$ everywhere on C	$f \in C^2$, C open convex	$\Leftrightarrow f$ convex
$\nabla^2 f(x) \geq 0$ at one x	inconclusive	counterexample $-x^4$ at 0
the first two are different statements, and only the third needs both hypotheses		

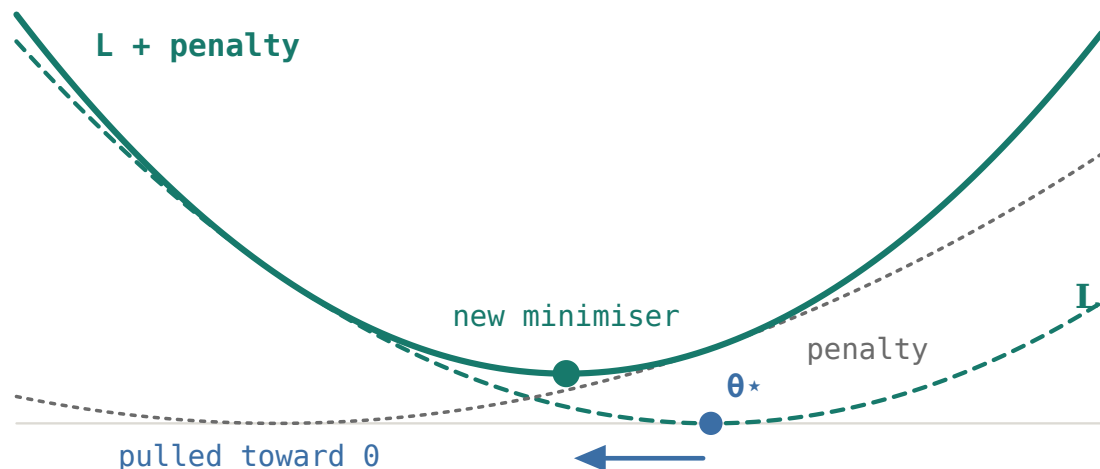
$\nabla f(x) = 0$ at an interior local min vs $\nabla f(x) = 0 \Rightarrow$ global min

left differentiable, interior;
necessary only

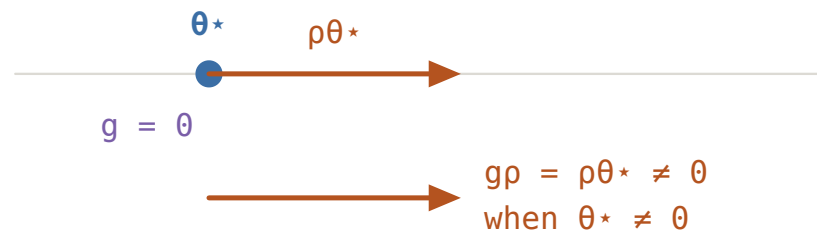
right adds convexity; now
sufficient and global

Convexity and Optimality

A quadratic penalty changes the problem



at θ^* , where L is lowest



so θ^* is no longer stationary, unless $\theta^* = 0$

$$L_\rho = L + \frac{\rho}{2} \|\theta\|^2,$$

$$g_\rho = g + \rho\theta, \quad \rho > 0$$

L_ρ L plus a second term

g_ρ the new gradient: $\nabla(\|\theta\|^2/2) = \theta$

$\rho > 0$ the penalty weight; chosen, not learned

θ^* where $g = 0$; if $\theta^* \neq 0$, $g_\rho \neq 0$

A new objective, so in general a new minimiser.

Convexity and Optimality

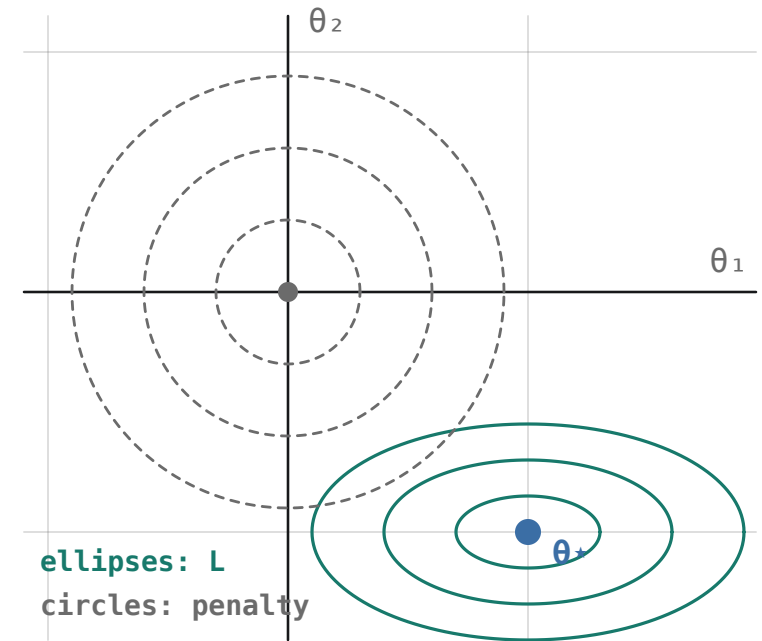
Certifying a quadratic solution

$H = \text{diag}(2, 8)$, $\theta^* = (1, -1)$, $\rho = 2$. Minimizer of L_ρ ?

(A) $(1, -1)$

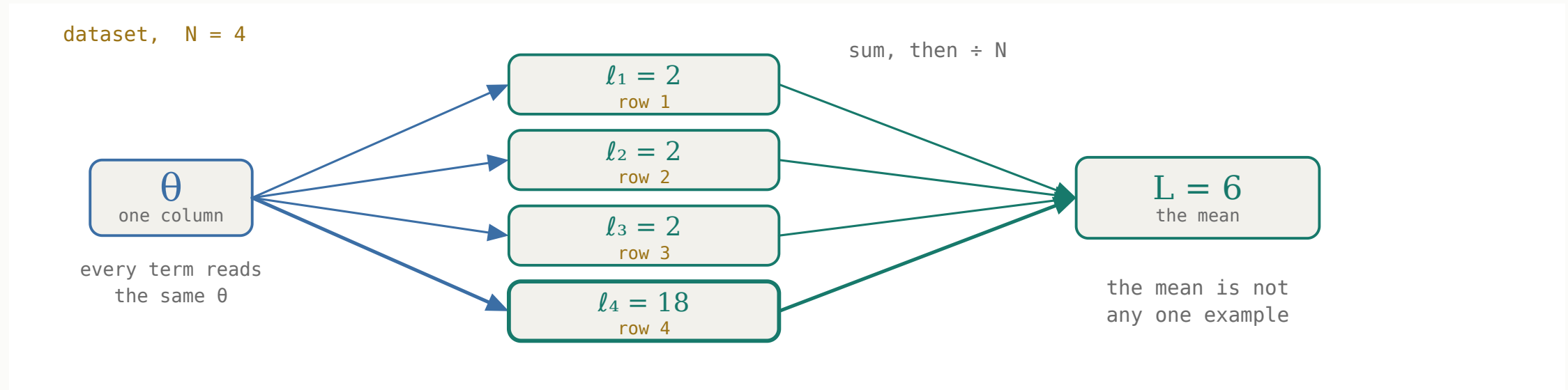
(B) $(\frac{1}{2}, -\frac{4}{5})$

(C) $(0, 0)$



Stochastic Gradients and Training

One dataset, many local losses



$$L(\theta) = \frac{1}{N} \sum_{i=1}^N \ell_i(\theta)$$

N examples in the set, here 4 $\ell_i(\theta)$ the loss on example i alone

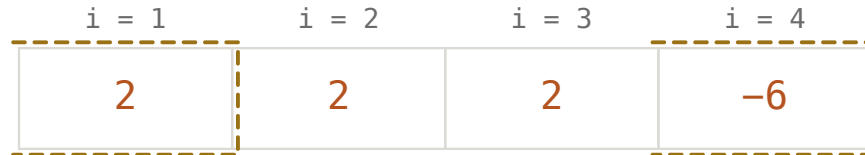
θ read by every term

L one number, whatever N is

Stochastic Gradients and Training

A mini-batch estimates the full gradient

$\nabla \ell_i$ at $\theta = 2$, one per example



drawn this step: i = 1 and 4

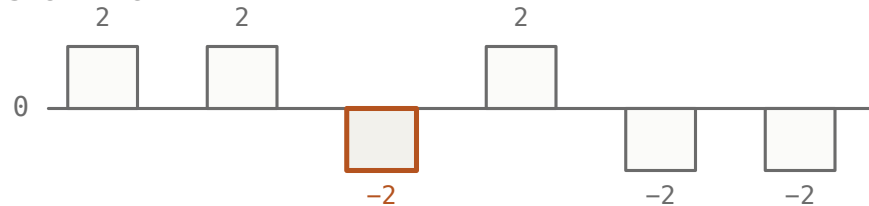
mean

$\hat{g} = -2$
this draw

$\nabla L = 0$
what we want

estimate is not the gradient

all six draws of two



not one of the six is ∇L
so a batch gives an estimate

$$\hat{g}_t = \frac{1}{B} \sum_{i \in \mathcal{B}_t} \nabla_{\theta} \ell_i(\theta_t)$$

$\hat{g}_t$ the estimate, shaped like θ $\mathcal{B}_t$ the indices drawn for step t

B how many, with $B \leq N$ θ_t the same in every term

Drawn uniformly without replacement from $\{1, \dots, N\}$, independently of θ_t , afresh each step.

Stochastic Gradients and Training

Add gradients before updating

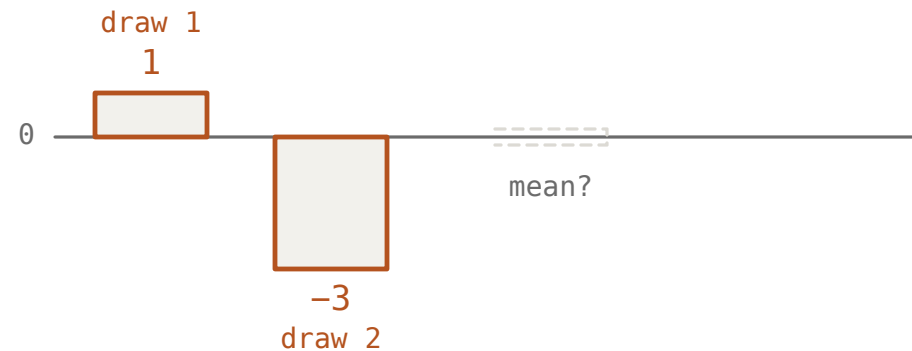
Sampled gradients 1 and -3 , with $B = 2$. Under the mean-loss update, where does θ go?

(A) $\theta - 2\eta$

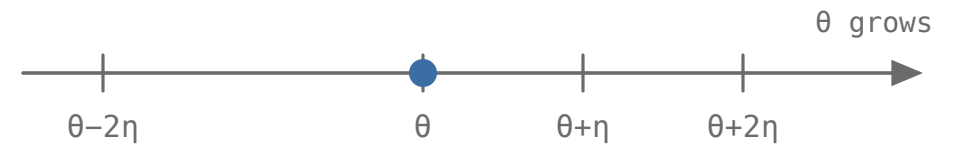
(B) $\theta + \eta$

(C) $\theta + 2\eta$

two sampled gradients



the parameter line



Stochastic Gradients and Training

Why uniform sampling has the right average

four one-example gradients

i = 1	i = 2	i = 3	i = 4
2	-4	1	5

$\nabla L = 1$
their mean

the full-data gradient

all 4 single draws, $w = 1/4$

draw i = 1 → 2
draw i = 2 → -4
draw i = 3 → 1
draw i = 4 → 5

average → 1

all 6 draws of two, $w = 1/6$

-1 3/2 7/2 -3/2 1/2 3

average → 1

not one of the six is 1
that gap is the noise

a nonuniform draw instead, $w = 7/10, 1/10, 1/10, 1/10$ → 8/5

uniform is what makes the two averages agree

$$\sum_{i=1}^N w_i \nabla \ell_i(\theta) = \nabla_{\theta} L(\theta) \quad \text{when } w_i = \frac{1}{N}$$

w_i how often index i is drawn

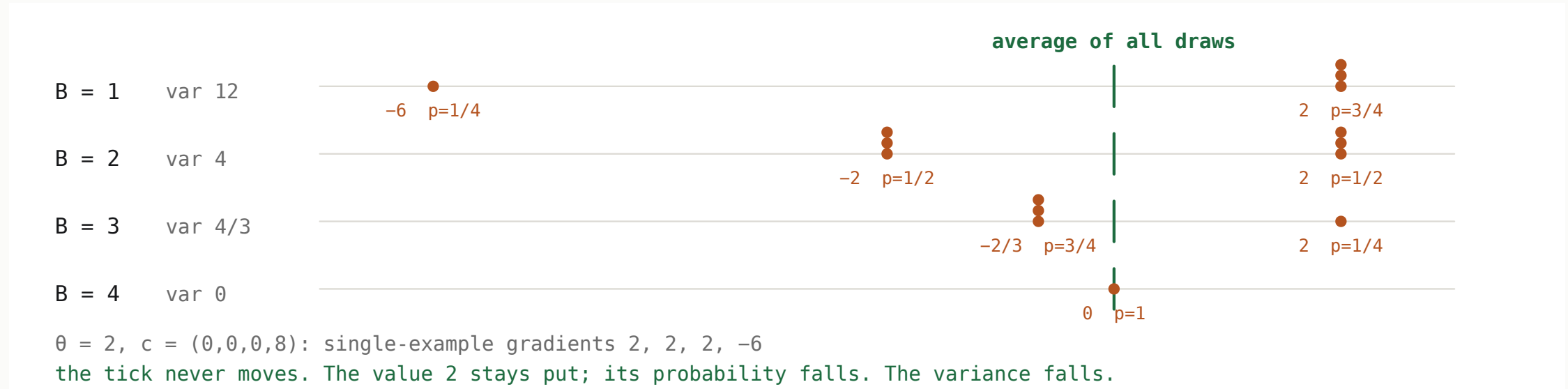
$\nabla \ell_i$ example i 's own gradient

∇L the full-data gradient

$w_i \neq \frac{1}{N}$ may change the average

Stochastic Gradients and Training

Unbiased at every B ; only the spread changes



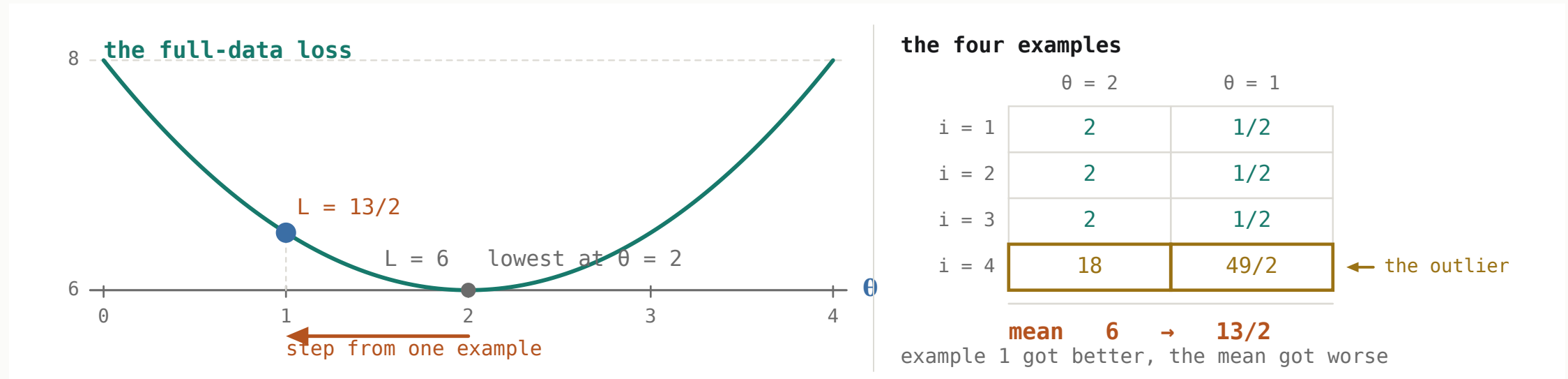
$$\mathbb{E}[\hat{g}_t \mid \mathcal{F}_t] = \nabla L(\theta_t), \quad \text{Var}(\hat{g}_t \mid \mathcal{F}_t) = \frac{\sigma^2}{B} \cdot \frac{N - B}{N - 1}$$

$\mathbb{E}[\cdot \mid \mathcal{F}_t]$ average over batches, given the history; a fresh uniform draw σ^2 population variance; scalar gradients here

$\frac{N-B}{N-1}$ without replacement, $N > 1$;
 0 at $B = N$

Stochastic Gradients and Training

A noisy step can raise the full loss



$$\ell_i(\theta) = \frac{1}{2}(\theta - c_i)^2, \quad c = (0, 0, 0, 8)$$

c_i example i 's own lowest point L the mean of the four

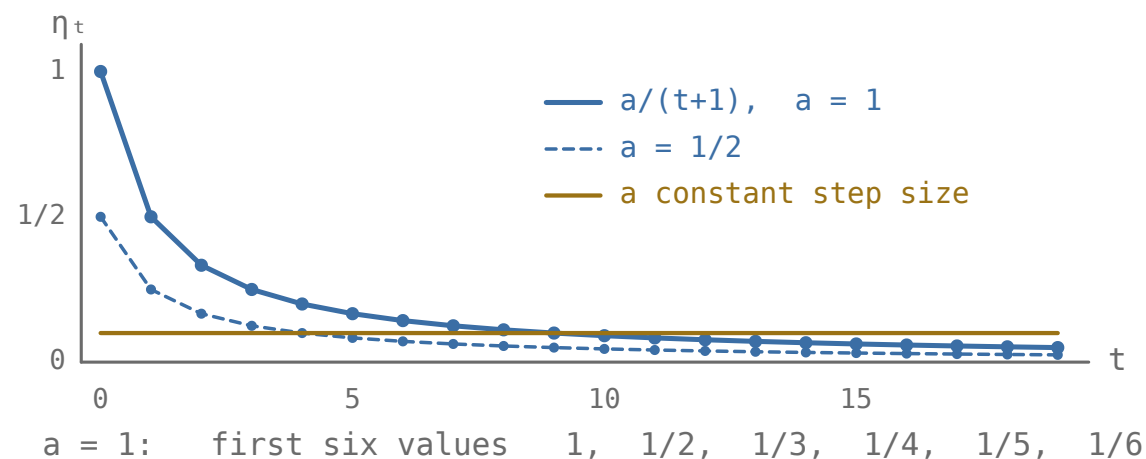
$\nabla \ell_1$ one example's gradient

WORKED EXAMPLE

$\nabla L(2) = 0$, so $\theta = 2$ minimises L , at $L = 6$. Draw example 1, with $\eta = \frac{1}{2}$: $\theta' = 1$ and $L = \frac{13}{2}$.

Stochastic Gradients and Training

First condition: no ceiling fixed in advance



what it says

the budget of step size never runs out, so the bound $G\sum\eta_t$ stops giving one fixed ceiling

what it does not say

η_t moves $\eta_t\|\hat{g}_t\|$, not η_t , so $\sum\eta_t$ bounds a distance only once $\|\hat{g}_t\| \leq G$

it does NOT make the trajectory unbounded, and does not promise the run leaves any ball:
if every $\hat{g}_t$ used is 0, θ never moves at all

$$\sum_{t \geq 0} \eta_t = \infty$$

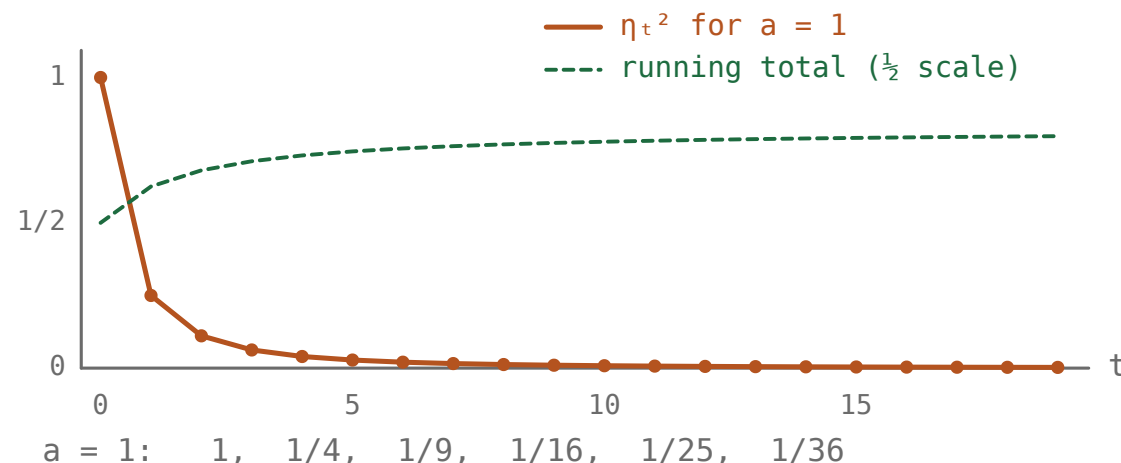
η_t step size at step t , chosen before training

needs $\|\hat{g}_t\| \leq G$ before it bounds any distance

$\sum \eta_t$ a sum of step sizes, not a distance travelled

Stochastic Gradients and Training

Second condition: the injected noise settles



what it says

noise contribution: $-\eta_t(\hat{g}_t - g_t)$
under the assumptions below, its signed cumulative sum converges; not a bound on path length or proof of training

what it needs

the error centred given the past, and its conditional second moment bounded
both series are hypotheses, not the theorem

$$\sum_{t \geq 0} \eta_t^2 < \infty$$

$\hat{g}_t - g_t$ the step's error; need not shrink on its own

needs the error centred, and its second moment bounded

$\sum \eta_t^2$ how fast the steps shrink, so the error's effect does

Stochastic Gradients and Training

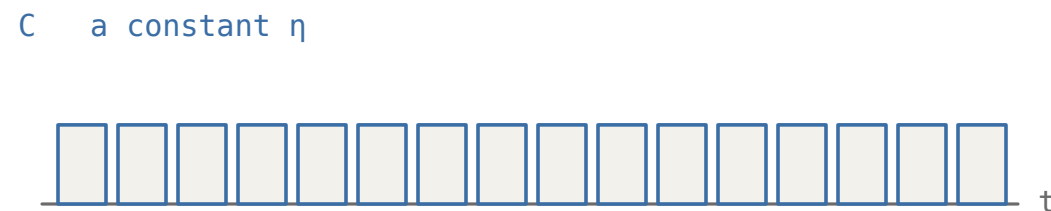
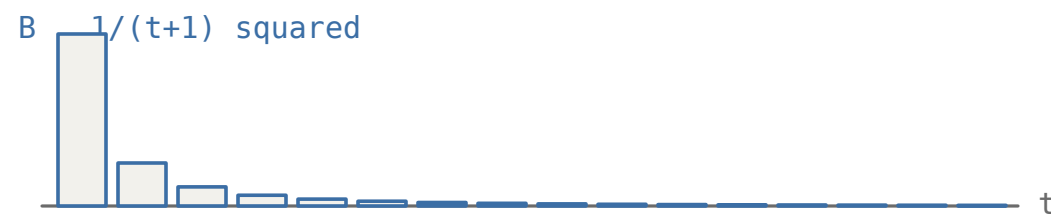
Which schedule passes both series tests?

Which η_t passes both $\sum \eta_t = \infty$ and $\sum \eta_t^2 < \infty$?

(A) $1/(t+1)$

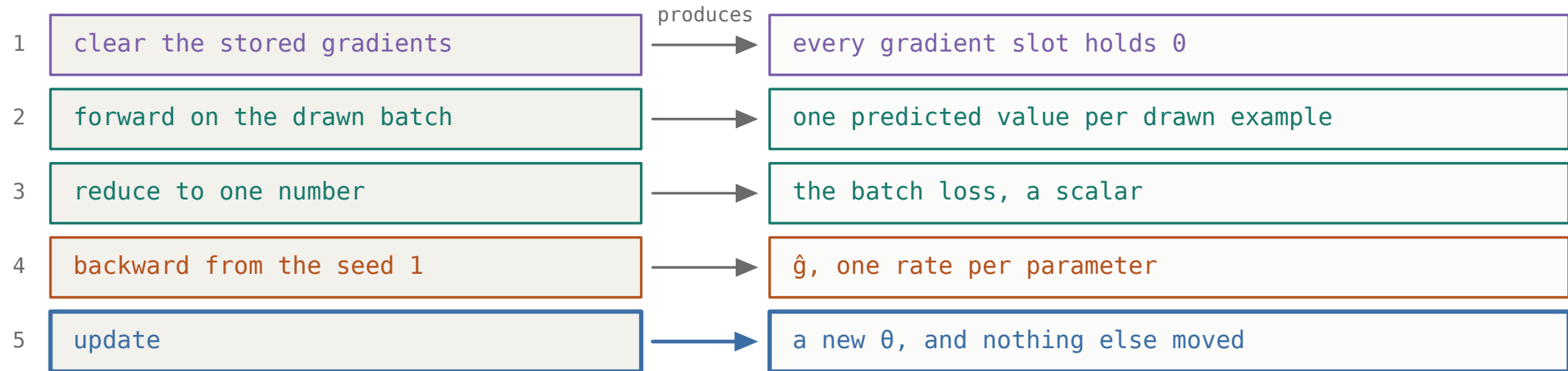
(B) $1/(t+1)^2$

(C) a constant $\eta > 0$



Stochastic Gradients and Training

One complete training iteration



$$\theta_{t+1} = \theta_t - \eta_t \hat{g}_t$$

θ_t the column before the step $\hat{g}_t$ what backward produced

η_t the schedule's size at t

Five operations, five different objects, and only the last one assigns to θ .

Stochastic Gradients and Training

Accumulation inside one graph: required

one graph, x used three times

$r = x * x + x, \quad x = 2$
| | |

2
left factor

+

2
right factor

+

1
direct use



5
dr/dx

every use contributes one term; the slot adds them

the += is the chain rule here, and is required

$$\bar{u} += \bar{v} \frac{\partial v}{\partial u}$$

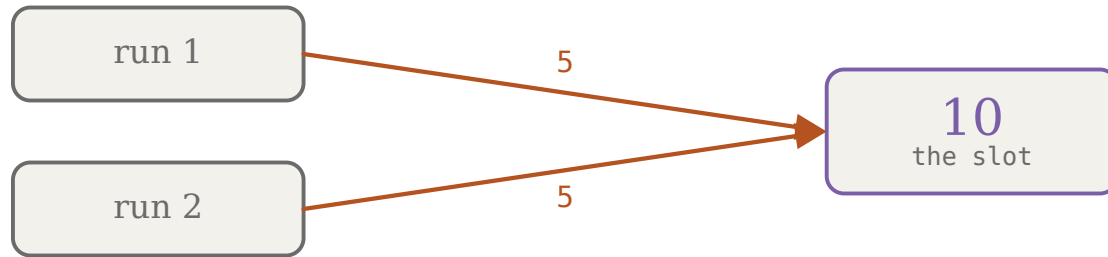
$\bar{u}$ the gradient slot for u
 $\frac{\partial v}{\partial u}$ the local factor, this use

$\bar{v}$ what arrived downstream
+= add every use; each term carries its own $\partial v / \partial u$

Stochastic Gradients and Training

Accumulation across two backward passes: a leftover

two fresh graphs, same θ , nothing cleared



$= \nabla(2L)$, not ∇L
correct, but of $2L$
each run alone would give 5

a leftover, and a bug unless you asked for it

$$\text{slot} = \hat{g}_1 + \hat{g}_2 = 10, \quad \hat{g}_1 = \hat{g}_2 = 5$$

$\hat{g}_i$ run i 's gradient, both at the
same θ

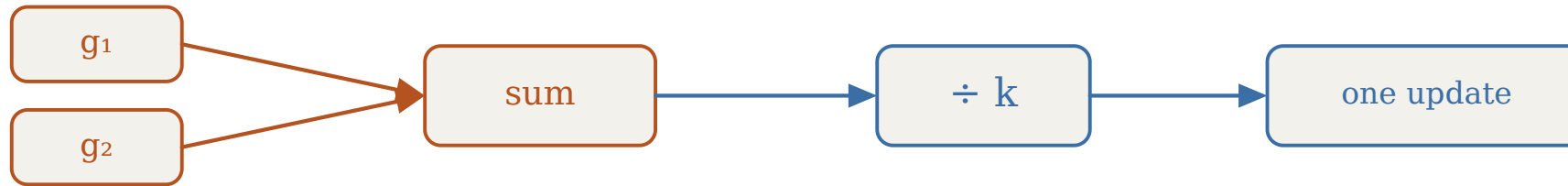
slot never cleared between the
two

Same arithmetic as the page before, different scope.

Stochastic Gradients and Training

Accumulation on purpose: micro-batches

k micro-batches at one θ , one update



equal sizes: $\div k$ gives the mean; plain SGD can instead use η/k

sizes 1 and 3, means 0 and 4: $\div k$ gives 2, the union mean is 3

a policy, not a mistake

$$\frac{\sum_m B_m \hat{g}_m}{\sum_m B_m} \xrightarrow{B_m \equiv b} \frac{1}{k} \sum_{m=1}^k \hat{g}_m$$

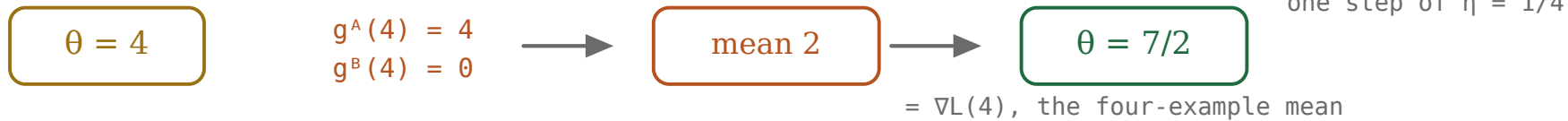
$\hat{g}_m$ micro-batch m 's mean
gradient

B_m its size; equal sizes collapse
the weights

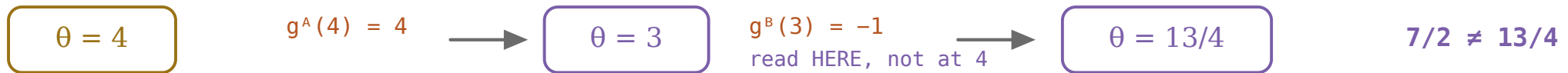
Stochastic Gradients and Training

Accumulate once, or step per micro-batch?

accumulate, then one step



step on each micro-batch



$$\theta - \eta \frac{g_A(\theta) + g_B(\theta)}{2} \neq (\theta - \eta g_A(\theta)) - \eta g_B(\theta - \eta g_A(\theta))$$

in general

left both read at θ ; one step on the full-data gradient

right g_B read after the first step, so somewhere else

Stochastic Gradients and Training

Locate the training-loop bug

Gradients are nonzero, yet θ never changes.
Which operation is the loop missing?

- ☐ (A) the update
- ☐ (B) a second backward
- ☐ (C) a larger batch

1. clear the gradient store
2. forward on the batch
3. form the loss
4. backward

5. ?

θ before

2.00

=

θ after

2.00

the gradient store

2

these two are equal, and the store is not zero

Stochastic Gradients and Training

A different omission: clearing, over two steps

both loops update, $\eta = 1/2$, θ starts at 2

	store at step 1	θ after step 1	store at step 2	θ ends
clearing	2	1	-7	9/2
never clearing	2	1	-5	7/2

both read their second gradient at $\theta = 1$, because both moved there

the store is what differs: -7 alone, or -5 with the leftover 2 still in it

$$\text{slot}_t = \sum_{s \leq t} \hat{g}_s \quad \text{instead of} \quad \hat{g}_t$$

$\hat{g}_s$ read at the θ the loop is
standing on

slot a running sum when line one
is missing

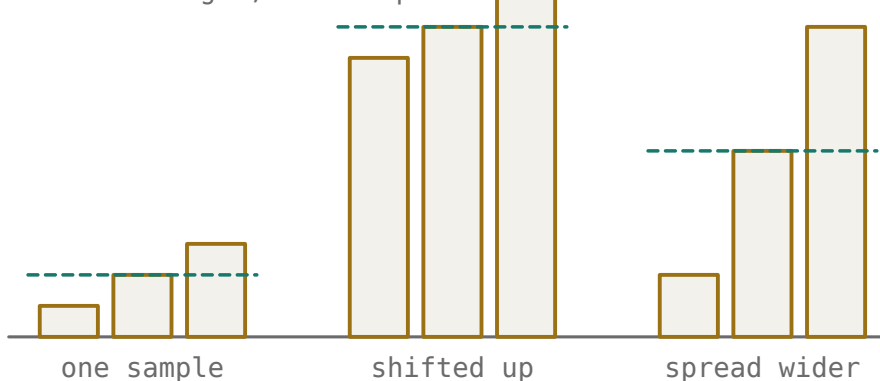
Both loops update. Only line one differs, and they end apart.

Layer Normalization

Controlling feature offset and scale

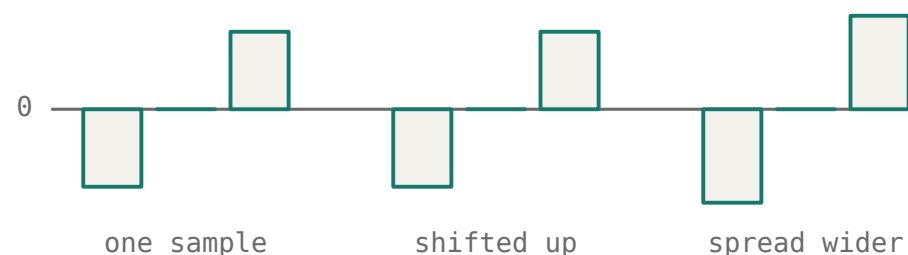
as they arrive

three averages, three spreads



centred, then divided by its own spread

now the three are comparable



$$h \in \mathbb{R}^d \mapsto z \in \mathbb{R}^d \quad \text{one sample at a time}$$

h one sample's feature column, $d \times 1$

d how many features, never how many samples

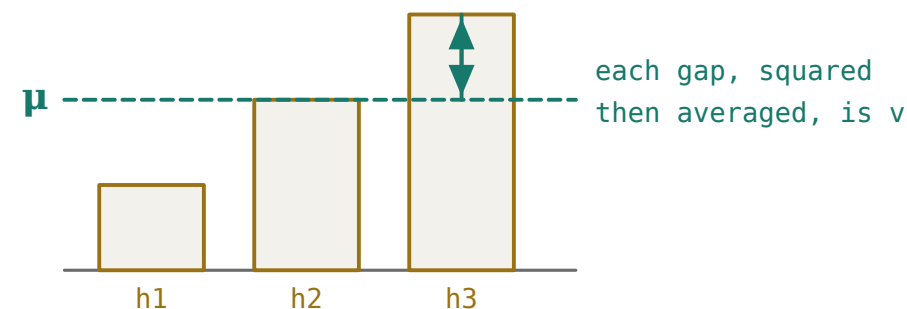
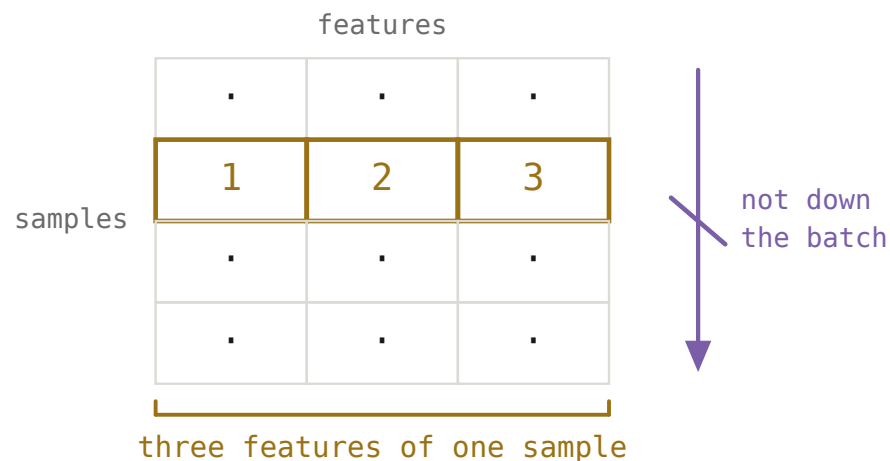
z what the next layer reads, same shape

two per feature a learned scale and shift, kept at the end

It controls feature offset and scale; faster training is not guaranteed.

Layer Normalization

Mean and variance over features



$$\mu = \frac{1}{d} \sum_{j=1}^d h_j, \quad v = \frac{1}{d} \sum_{j=1}^d (h_j - \mu)^2$$

h_j feature j of this one sample

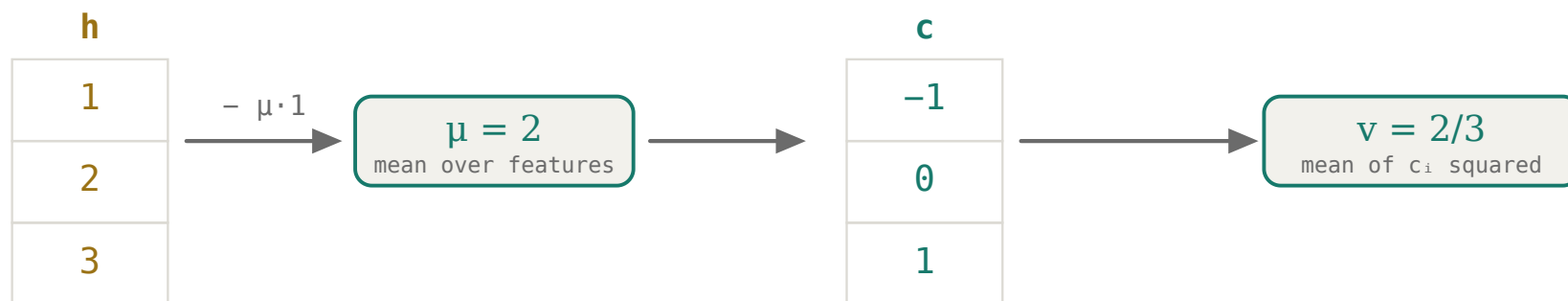
μ, v two scalars, built from this sample alone

$j = 1, \dots, d$ the sum runs over features; no sample index appears in it

Divided by d , not $d - 1$: the population variance of the d numbers in front of us.

Layer Normalization

Step one: centre the features



$-1 + 0 + 1 = 0$: centring always gives a column that sums to zero one sample at a time. Nothing here looks at any other sample.

$$\mu = \frac{1}{d} \sum_i h_i, \quad c = h - \mu \mathbf{1}, \quad v = \frac{1}{d} \sum_i c_i^2$$

h one sample's features, $d \times 1$

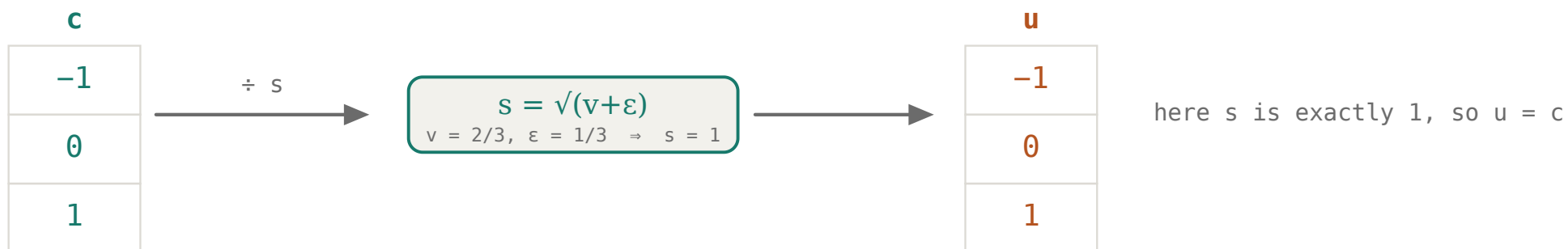
μ mean over features, not over the batch

c the centred features; $\sum_i c_i = 0$ always

v their mean square, this sample's spread

Layer Normalization

Step two: divide by s , and why ε is there



$\varepsilon > 0$ keeps s away from 0 when every feature of a sample agrees
and it is why $\text{mean}(u^2) = v/(v+\varepsilon)$ is below 1, not equal to it

$$u = \frac{c}{s}, \quad s = \sqrt{v + \varepsilon}, \quad \varepsilon > 0$$

s one positive number for this sample

ε fixed, so s is never 0 on a flat sample

u the normalized features; their statistics are the next vote

Layer Normalization

Step three: the learned scale and shift



one component in full

$$\begin{aligned} z_1 &= \gamma_1 u_1 + \beta_1 \\ &= 2 \cdot (-1) + 0 \\ &= -2 \end{aligned}$$

centring loses μ ; learned γ and β cannot restore that missing mean

$$z = \gamma \odot u + \beta, \quad z_i = \gamma_i u_i + \beta_i$$

γ, β learned, $d \times 1$ each, one pair
for all samples

z what leaves the block, $d \times 1$

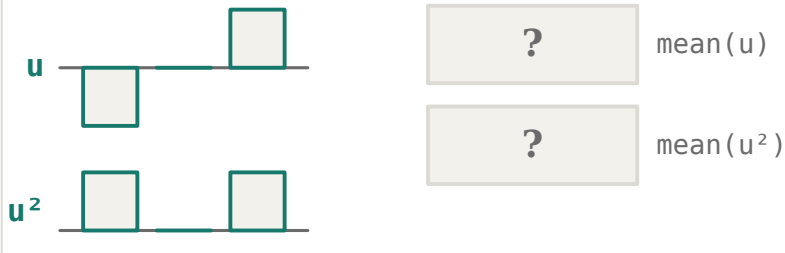
$\odot$ entry by entry, not a matrix
product

Layer Normalization

What is exactly zero or one?

$\varepsilon = 1/3$ here, not a library's $1e-5$:
enlarged on purpose, so the effect is
big enough to see.

$s = 1$, so $u = (-1, 0, 1)$



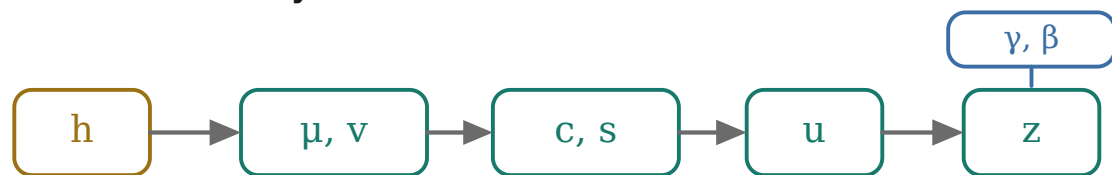
For a **nonconstant** h and $\varepsilon > 0$, the normalized u has

- (A) mean 0, $\text{mean}(u^2) = 1$
- (B) mean 0, and $\text{mean}(u^2) < 1$
- (C) no fixed mean

Layer Normalization

Where the backward pass is written down

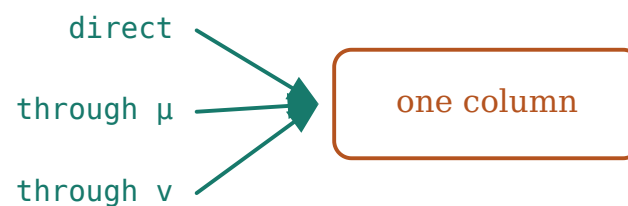
forward: already written



backward: the supplement

one column down, one for each of the learned pair

three routes in



two corrections out

$$\nabla_z L \mapsto \nabla_h L, \nabla_\gamma L, \nabla_\beta L$$

$\nabla_z L$ from above, $d \times 1$

$\nabla_h L$ back down; three routes build it

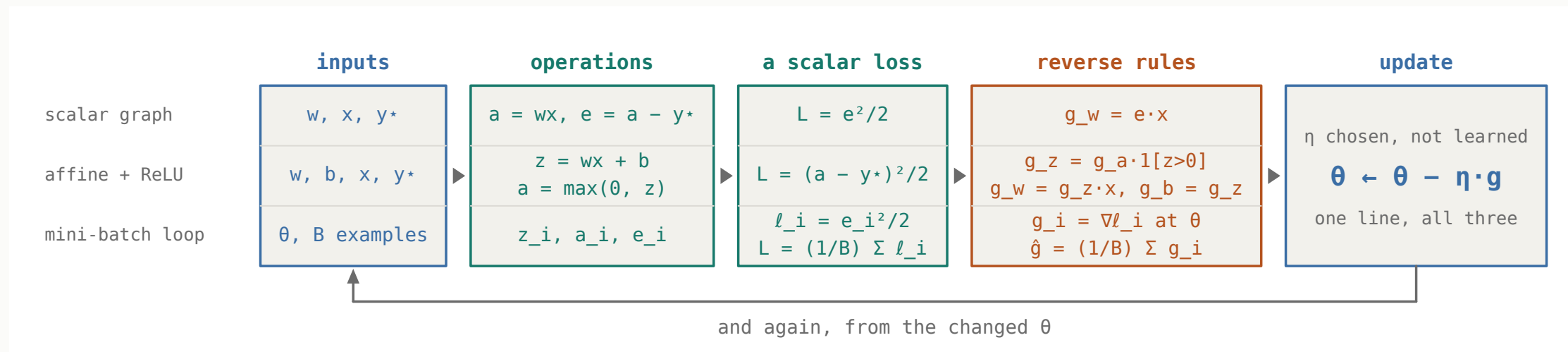
$\nabla_\gamma L, \nabla_\beta L$ $d \times 1$ each, like the pair

- two correction terms, and a zero-sum test that passes a wrong formula
- wanted again in Lecture 10, and by a gradient check

Eleven pages, with the worked numbers and its own three votes: jingz6676.github.io/rob6013/layernorm-backward.html

A Complete Learning Step

The same chain rule in every block



$$\bar{L} = 1, \quad \bar{u} += \bar{v} \frac{\partial v}{\partial u}, \quad \theta_{t+1} = \theta_t - \eta_t g_t$$

$\bar{L} = 1$ the only seed a reverse pass needs

$\partial v / \partial u$ one local rule, off saved values

$\bar{v}$ what arrived from downstream

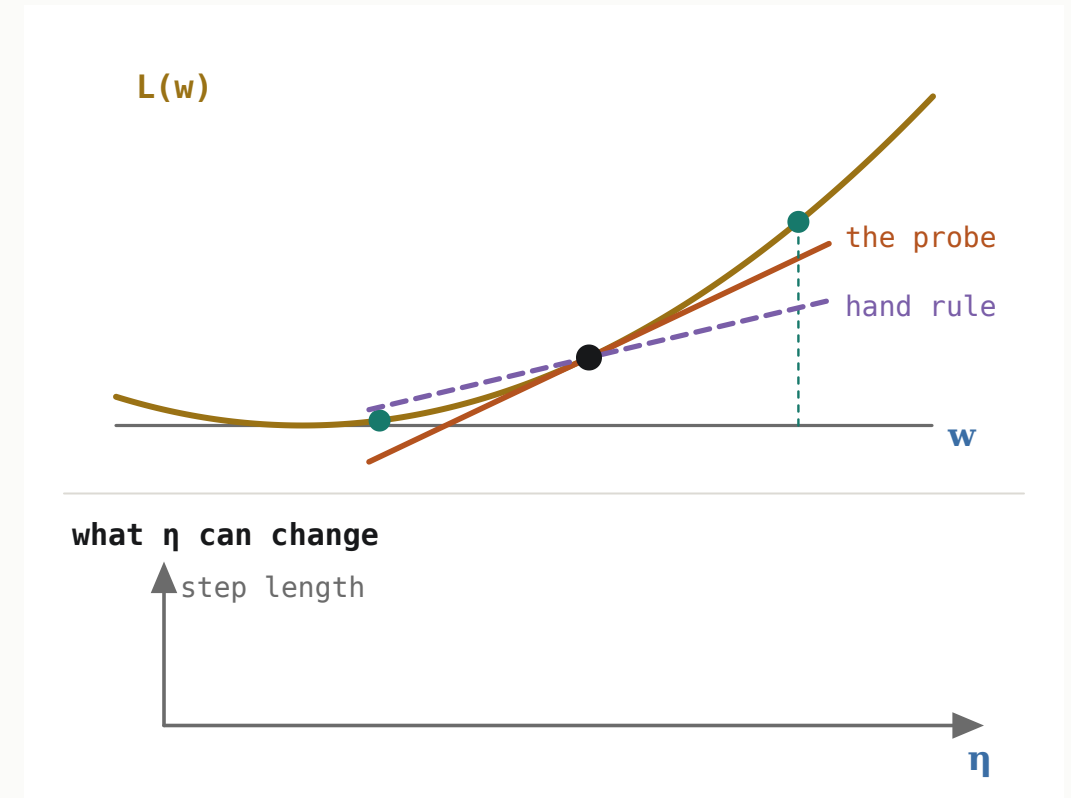
η_t enters here, nowhere earlier

A Complete Learning Step

Diagnose before changing the optimizer

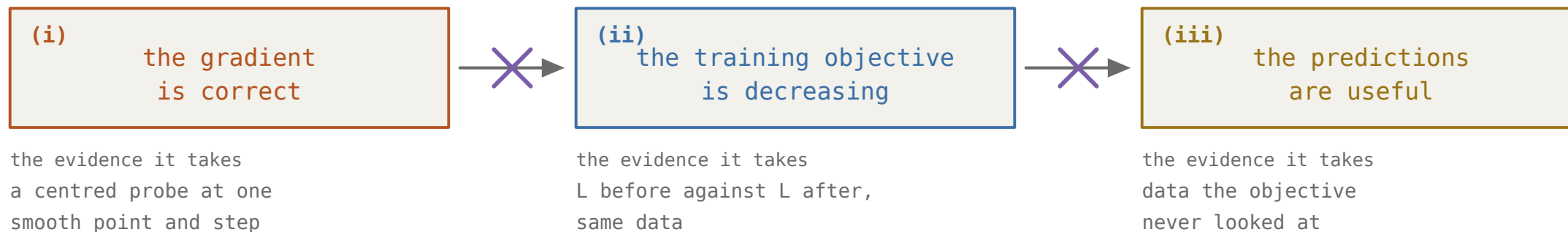
Loss rising; the hand rule and a smooth probe disagree. First action?

- (A) swap the optimizer
- (B) inspect derivative, fixture, reduction
- (C) raise η



A Complete Learning Step

Three distinct claims about training



neither arrow holds: each claim needs its own evidence

(i) $g = \nabla_{\theta} L$ (ii) $L(\theta_{t+1}) < L(\theta_t)$ (iii) ℓ at an unseen x

(i) a derivation gives the rule; a probe tests the code (ii) a step question; η and curvature decide it

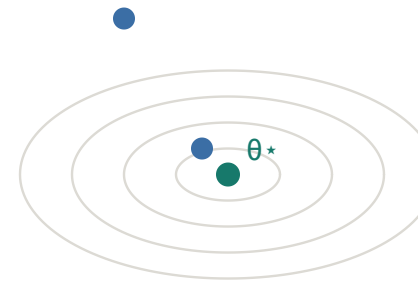
(iii) a data question, asked where L never looked

A Complete Learning Step

Exit check and next connection

Which guarantee was proved?

- (A) any neural network, $\eta < 2/\lambda_{\max}$ at the start
- (B) a fixed $H \succ 0$ quadratic, that bound
- (C) correct gradients, good test predictions



one fixed H

$$H = \text{diag}(2, 8)$$

$$\theta^* = (1, -1)$$

$$\eta = 1/5$$

three starts

multiplier $1 - \eta\lambda$, one per mode

inside the band, the mode contracts



$$H = 100 I, \text{ same } \eta: \lambda = 100$$

References

Where these rules come from

1. Boyd & Vandenberghe. [*Convex Optimization*](#), ch. 2–3, 9. Cambridge, 2004.
2. Goodfellow, Bengio & Courville. [*Deep Learning*](#), ch. 8, §8.3.1. MIT Press, 2016. Momentum and Adam, later in that chapter, are optional reading.
3. Baydin et al. [*Automatic Differentiation in Machine Learning: a Survey*](#). JMLR, 2018.
4. Ba, Kiros & Hinton. [*Layer Normalization*](#). arXiv, 2016.

References

Implementations, and the course standards

5. Vaswani et al. [*Attention Is All You Need*](#), §3.2.1. NeurIPS, 2017. The map a later lecture derives; reading ahead, not used here.
6. [*PyTorch autograd tutorial*](#). PyTorch documentation, read 2026-10-04.
7. Karpathy, A. [*micrograd*](#). GitHub. An implementation to read, and to license-check before copying.

Course standards: ROB-GY 6013, [*Notation and Conventions*](#) · columns, the gradient row, the batch layout. ROB-GY 6013, [*Textbooks and Reading Guide*](#) · what each source is for, and where its notation differs.